FES Stats Workshop

Overview of Stats process

SCAN TO REGISTER

REGISTRATION REQUIRED

Reserve your spot to attend

Bring a laptop. Open to all levels — basic stats and some R are helpful but not required.

forms.cloud.microsoft/r/gJfRw8wJd9 →

Materials and code:

Outline for today

8:00 – 8:30: General stats overview

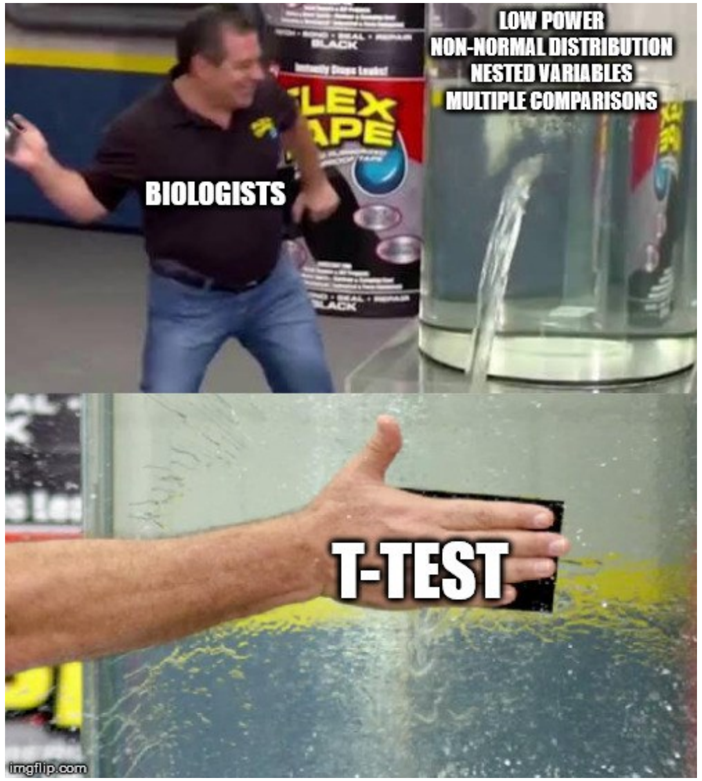
8:30-10:00: Survival analysis w/ examples in R

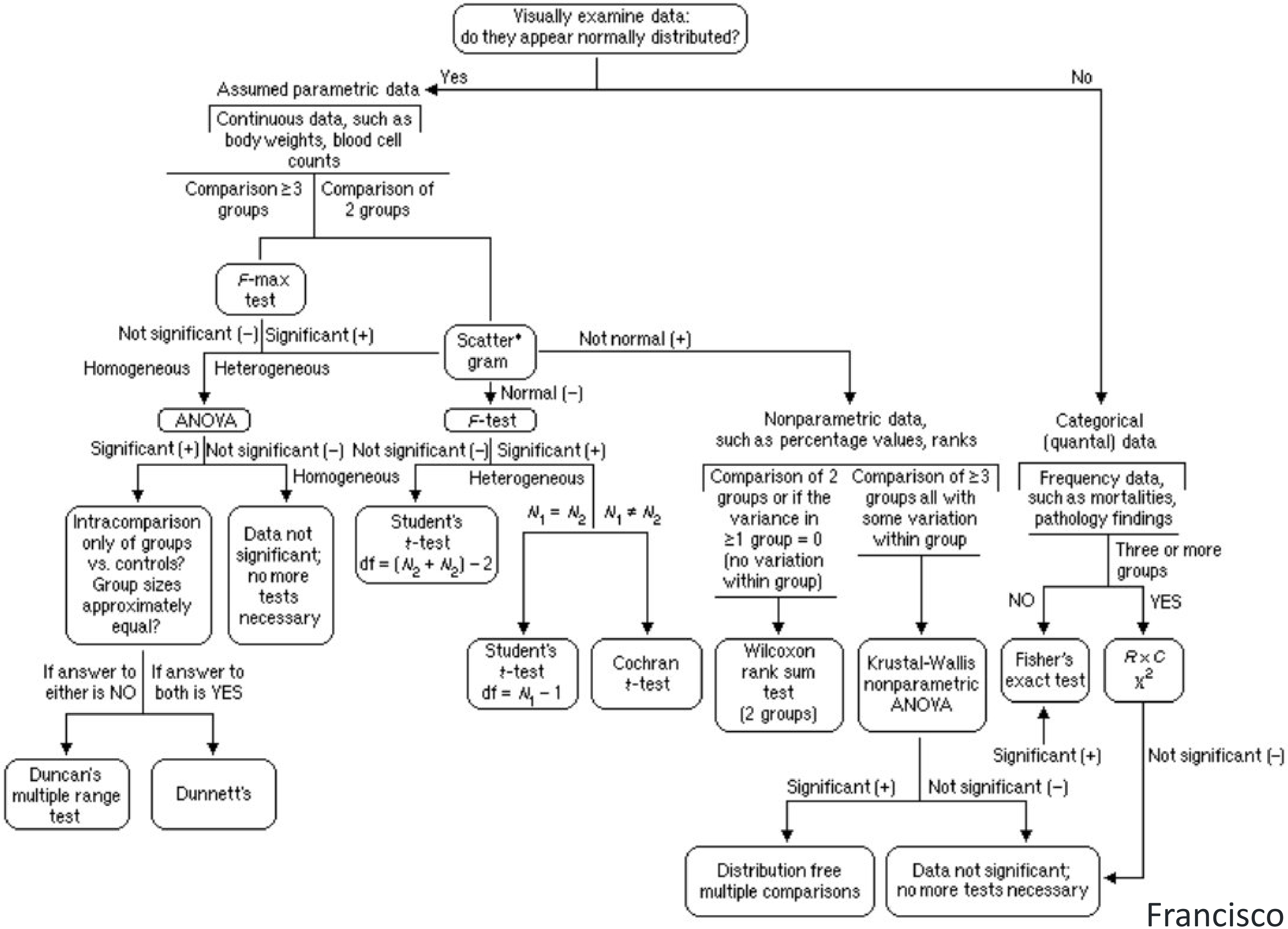
10:00-11:30: Dose-dependent w/ examples in R

11:30-12:00: Questions and general discussion

GET CODE HERE:

– https://github.com/hahnp13/Workshops.git

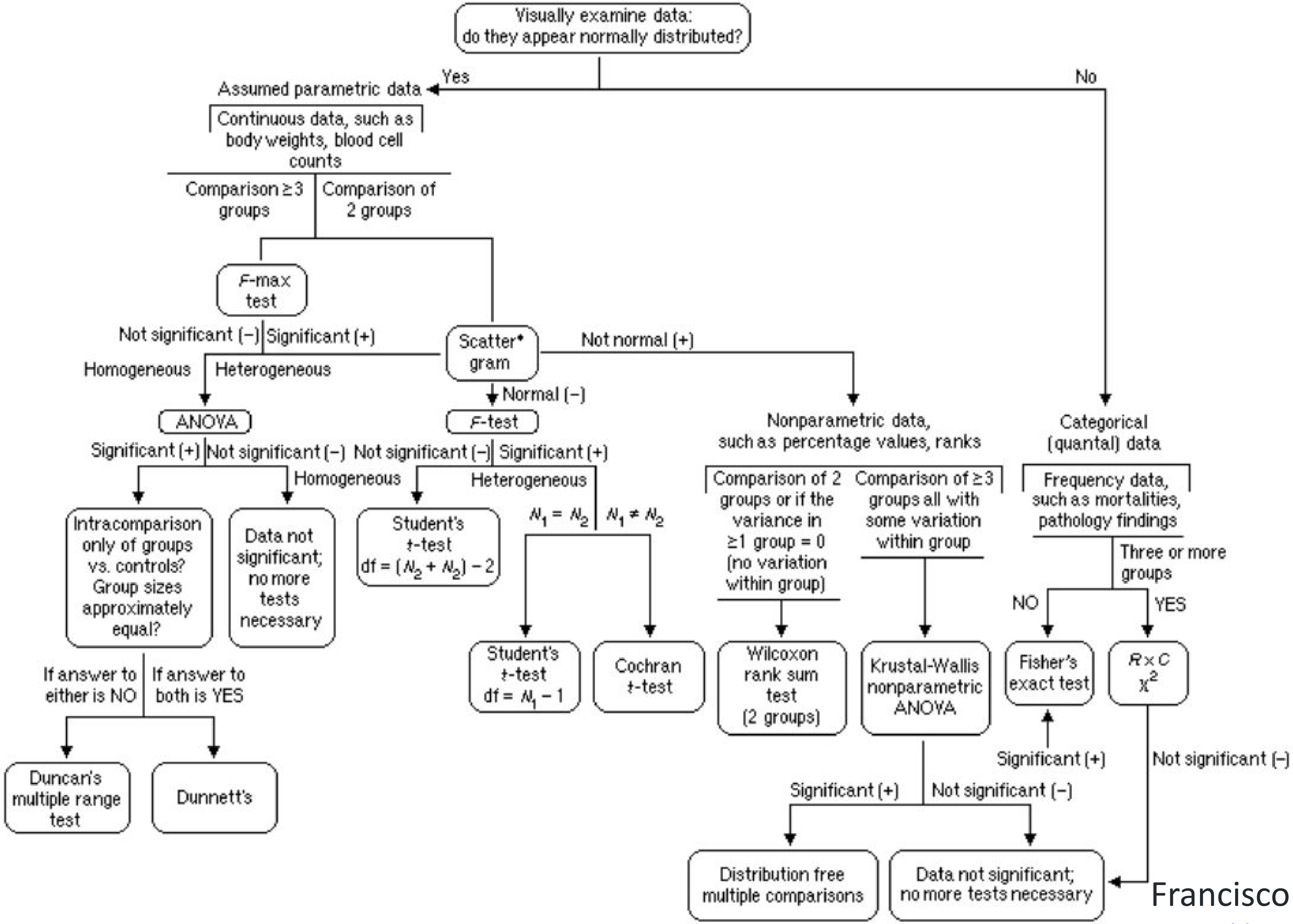




Francisco Rodriguez-Sanchez

https://frodriguezsanchez.net

Modern statistics are easier than this



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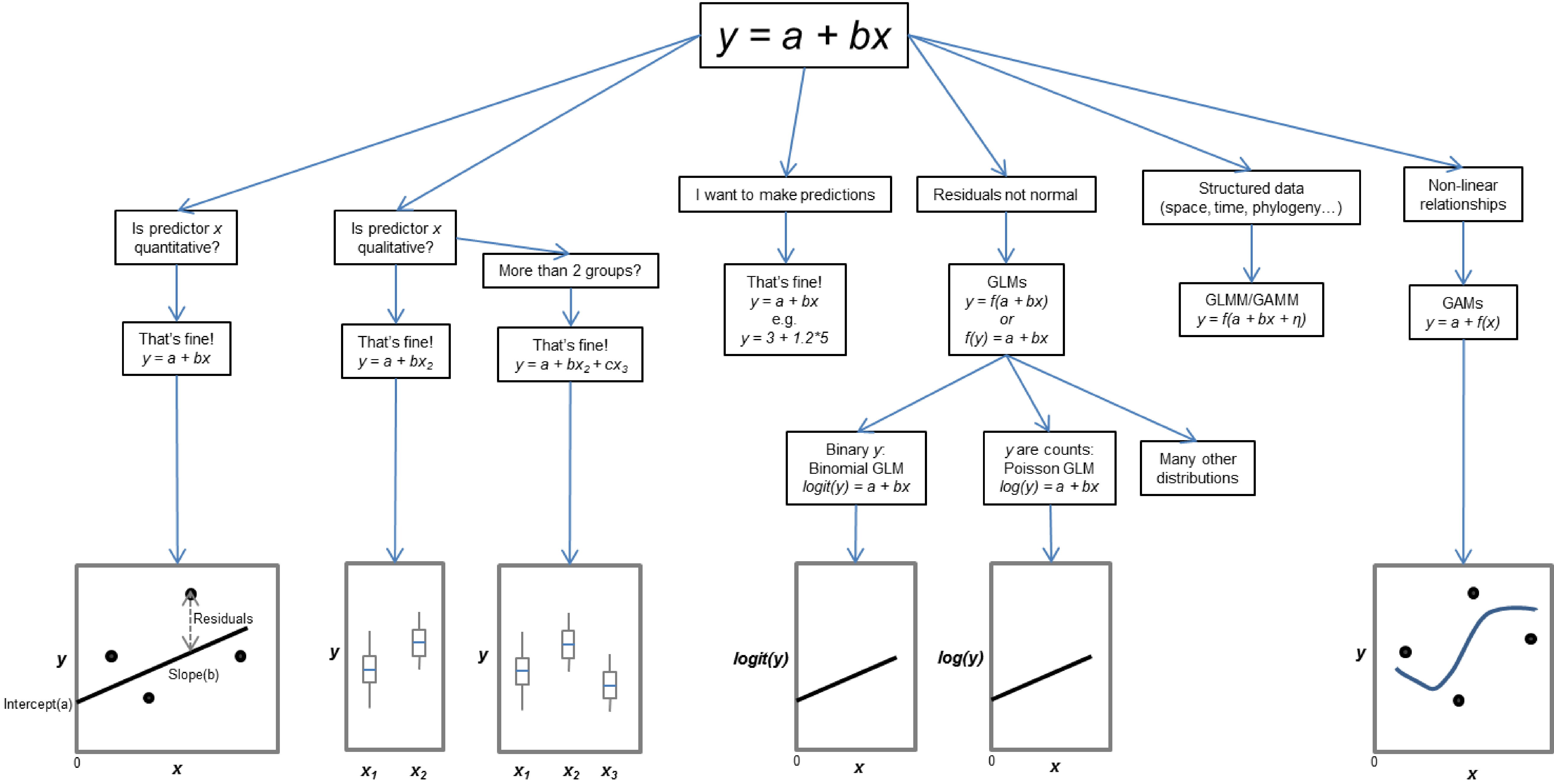

Francisco Rodriguez-Sanchez

https://frodriguezsanchez.net


QUIZ
TIME
makeagif.com

What is a probability distribution?

1. A measure of how spread out the data values are from the mean
2. A mathematical description of a random phenomenon that gives the probability of possible events
3. A summary statistic that describes the center of a data set
4. An estimate of the variability, or precision, of a sample mean

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What is a standard deviation?

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What is a standard error?

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What is a standard error?

1. A measure of how spread out the data values are from the mean
2. A mathematical description of a random phenomenon that gives the probability of possible events
3. A summary statistic that describes the center of a data set
4. **An estimate of the variability, or precision, of a sample mean**

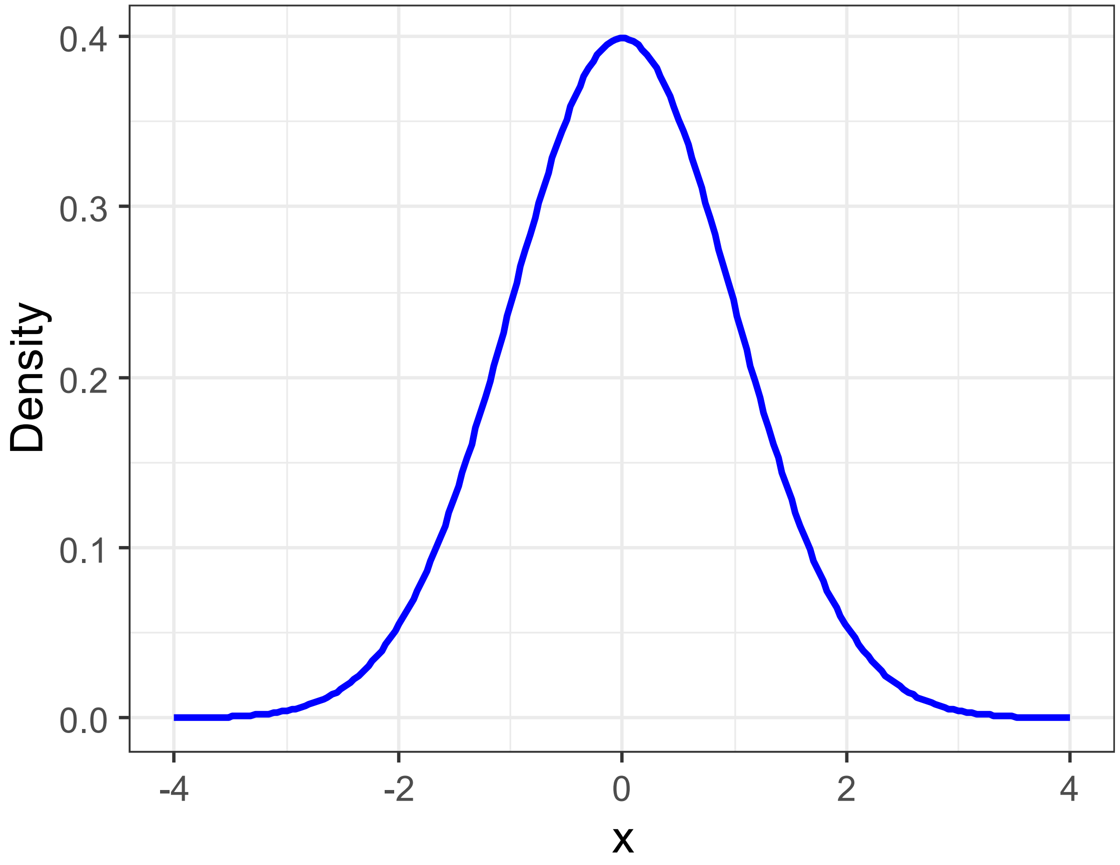
Questions ???

Materials and code:

What is a probability distribution?

Normal (Gaussian) distribution

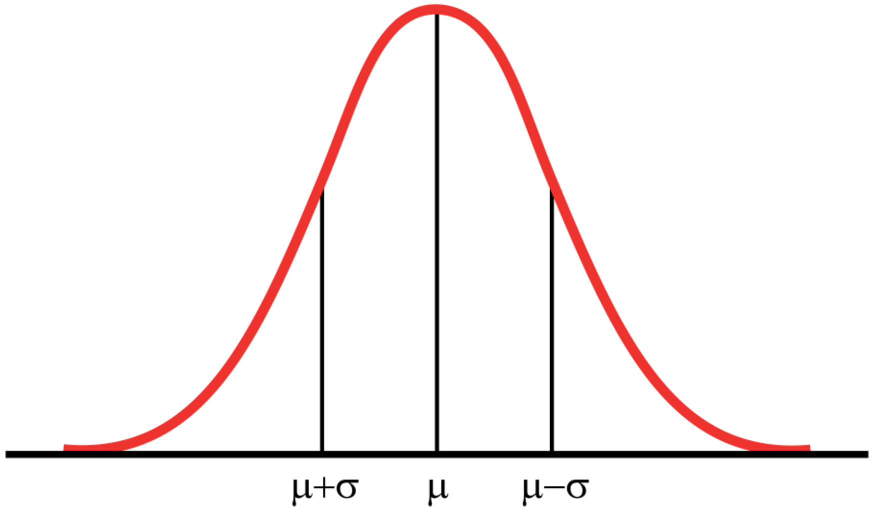
- Continuous numeric variables
  - -∞ to ∞
  - Most points near the mean
  - Points symmetrically distributed around mean

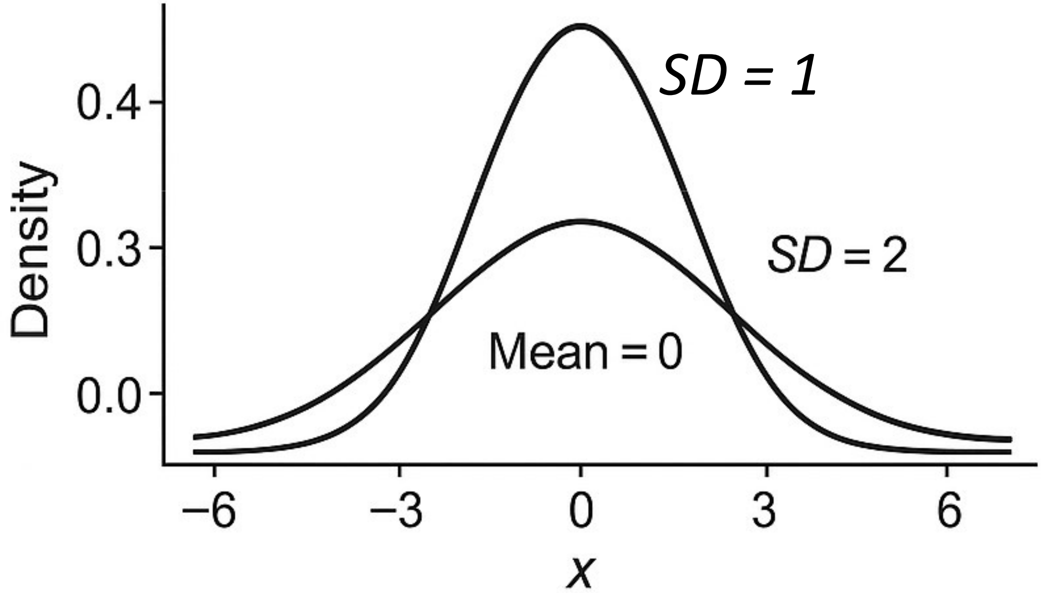


What is a probability distribution?

Normal (Gaussian) distribution

- Continuous numeric variables
  - -∞ to ∞
  - Most points near the mean
  - Points symmetrically distributed around mean
- Parameters
  - μ = Mean
  - σ² = Variance
    - σ = Standard deviation



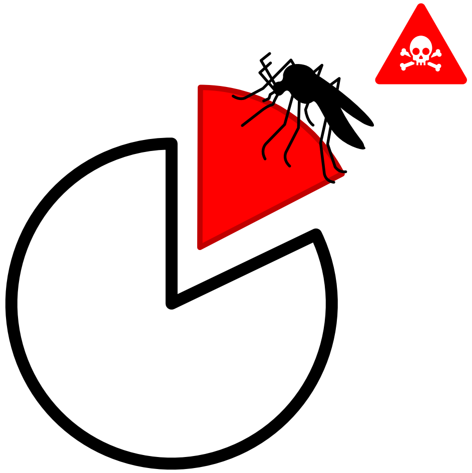


What processes generate data?

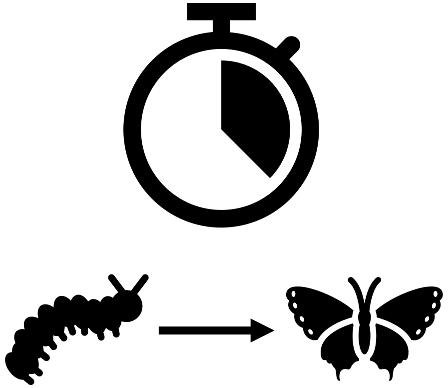
- Data are the result of natural (or biological) phenomena

Counts

Proportion



Time-to-Event

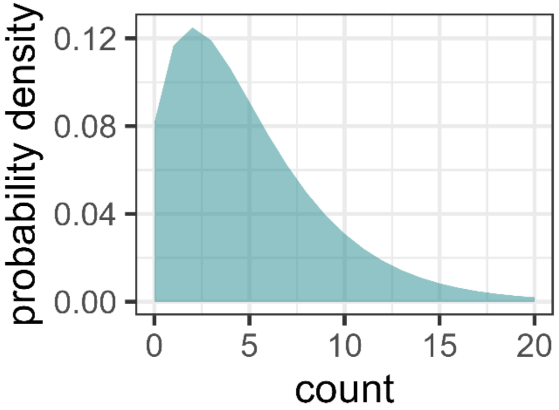


What processes generate data?

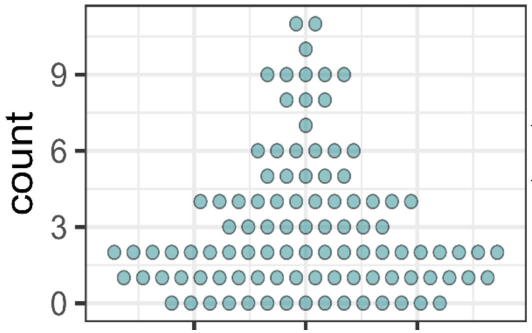
- Data are the result of natural (or biological) phenomena
- Phenomena is mathematically described by a probability distribution
- Data we observe are a random subset of events produced by a phenomenon
- Used to model data in statistical analysis and make inferences

Natural processes

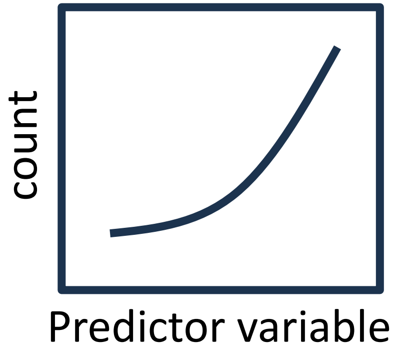
Probability distribution



Observed data



Model / inference



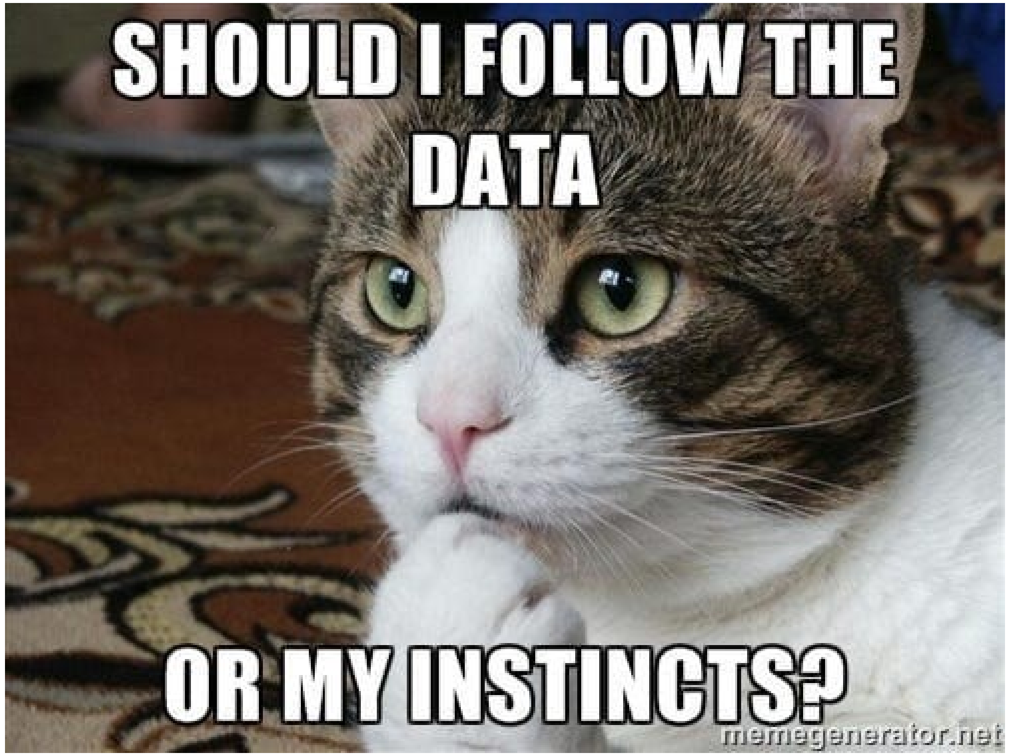
Steps for constructing and interpreting a GLMM

0. Data Exploration
1. Construct model
    a. Fixed effects (predictor or explanatory variables)
    b. Random effects (blocking variables)
    c. Error distribution (response variable or dependent variable)
2. Assess residuals and overdispersion
3. Check random effects and/or distribution using AIC (not always needed)
4. Examine significance of terms in model
    1. Anova
    2. AIC model selection
5. Calculate treatment means and/or slopes
6. Construct contrasts
7. Calculate R2m and R2c
8. Make pretty figure

Step 0: Data exploration

1. Formulate biological hypothesis
   Carry out experiment
   & collect data
2. **Data Exploration**
3. Apply statistical model

Use to choose an appropriate model
and avoid assumption violations



Methods in Ecology and Evolution

British Ecological Society

Methods in Ecology and Evolution 2016, 7, 636–645 doi: 10.1111/2041-210X.12577

SPECIAL FEATURE: 5TH ANNIVERSARY OF *METHODS IN ECOLOGY AND EVOLUTION*

A protocol for conducting and presenting results of regression-type analyses

Alain F. Zuur[1]* and Elena N. Ieno[2]

[1]Highland Statistics Ltd., 9 St Clair Wynd, Newburgh AB41 6DZ, UK; and [2]Highland Statistics Ltd., Av Finlandia 11, Bwg 34, Santa Pola, Spain

Step 0: Data exploration

1. Formulate biological hypothesis
   Carry out experiment
   & collect data
2. **Data Exploration**
3. Apply statistical model

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1. Understand design & variables
2. Explore data
   a. Values & Outliers
   b. Distribution & Variance
   c. Collinearity
   d. Relationships / Interactions
   e. Independence / Clustering
3. Document decisions

Methods in Ecology and Evolution 2010, **1**, 3–14 doi: 10.1111/j.2041-210X.2009.00001.x

A protocol for data exploration to avoid common statistical problems

Alain F. Zuur*[1,2], Elena N. Ieno[1,2] and Chris S. Elphick[3]

[1]*Highland Statistics Ltd, Newburgh, UK;* [2]*Oceanlab, University of Aberdeen, Newburgh, UK; and* [3]*Department of Ecology and Evolutionary Biology and Center for Conservation Biology, University of Connecticut, Storrs, CT, USA*

Step 1. Construct a statistical model to test your hypothesis

1. Construct a statistical model to test your hypothesis.
   – Linear model: y ~ x1 + x2 + x1:x2
   – Random effects?
   – Distribution? Gaussian okay or generalized model?

Step 1. Construct a statistical model to test your hypothesis

- Linear model: $y_i \sim \mu + \beta + e_i$

Response variable

mean

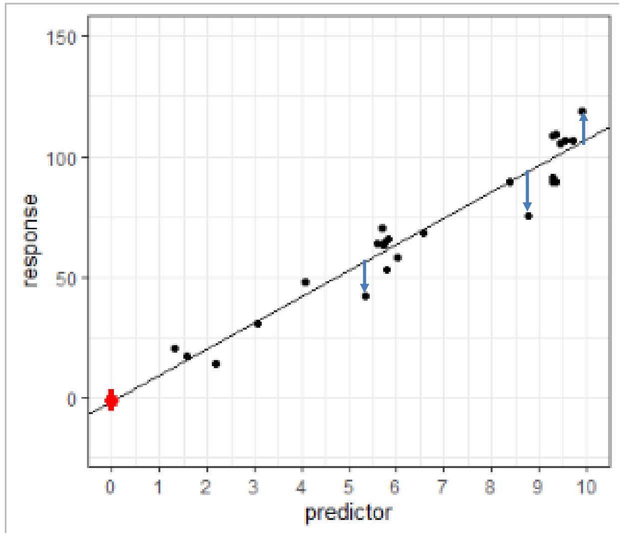
Parameter estimate to change the mean based on some variable (ie. slopes or intercepts)

Residual variation or error

Step 1. Construct a statistical model to test your hypothesis

- "Linear regression": $y_i \sim B_0 + B_1X + e_i$

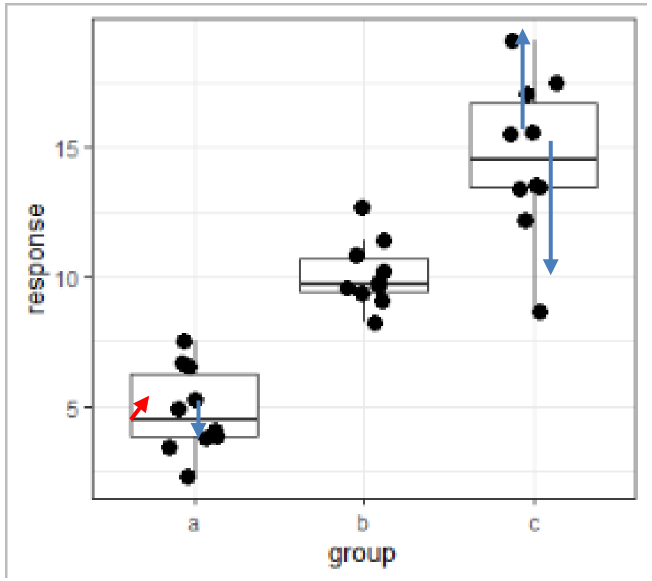
  intercept   slope   resid



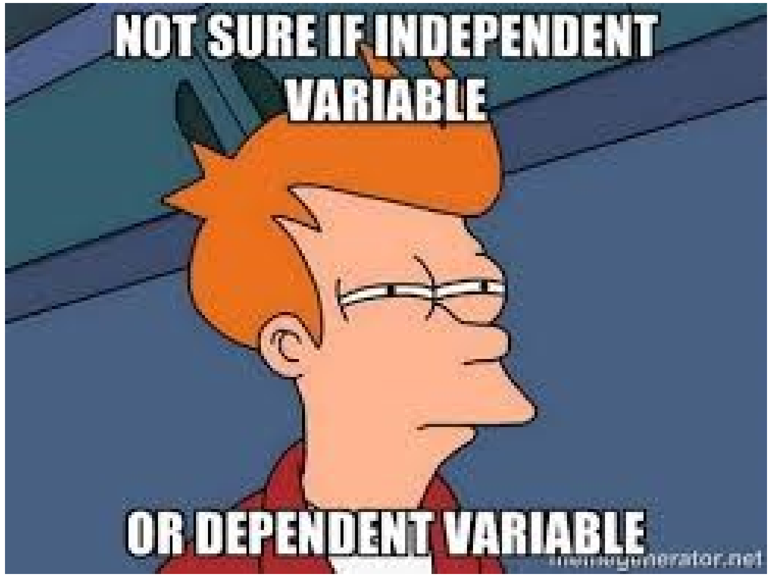
- "ANOVA" model: $y_{ij} \sim B_0 + \beta_j + e_{ij}$

  intercept   resid

  Adjustment to intercept for each group



Questions ???

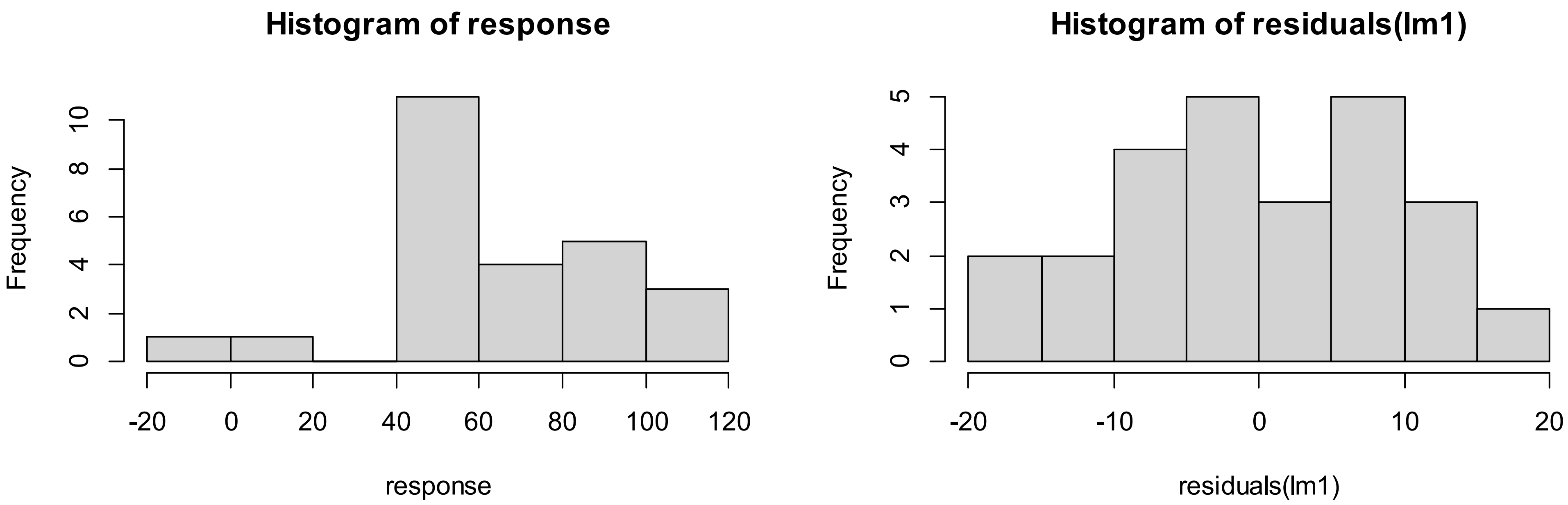


Materials and code:

Step 2. Check model assumptions

Assumptions of linear models

- **Response variable errors (ie. residuals) normally distributed**

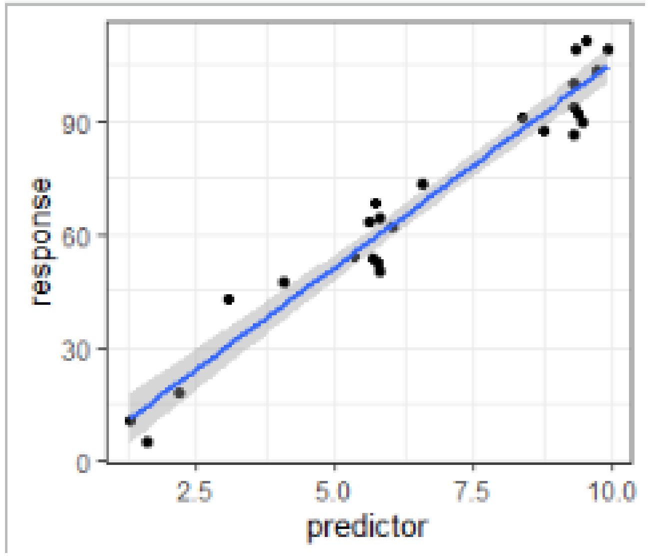


Residuals should be normally distributed

Assumptions of linear models

- Response variable errors (ie. residuals) normally distributed
- **Linearity**

Good



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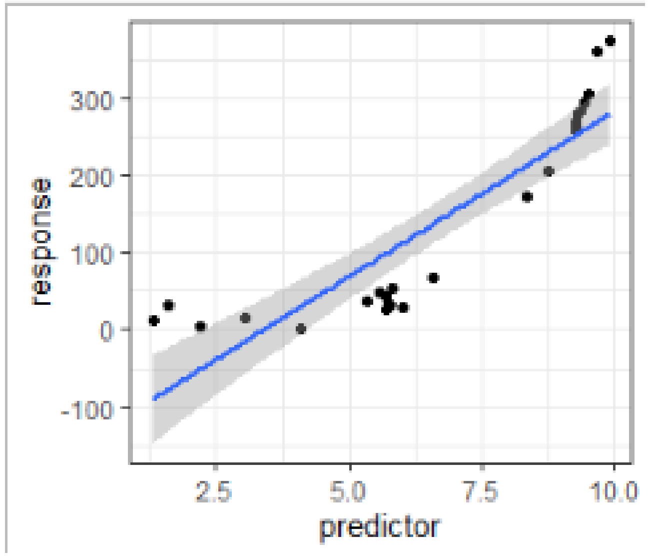
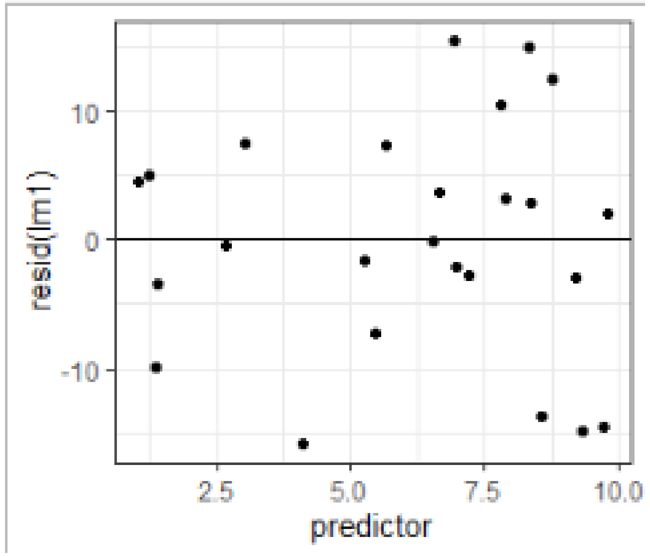


Assumptions of linear models

- Response variable errors (ie. residuals) normally distributed
- Linearity
- **Homoscedasticity**

Good



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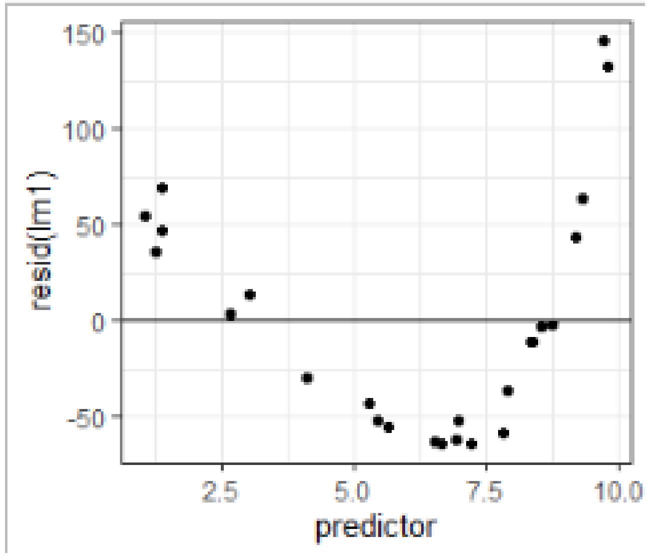
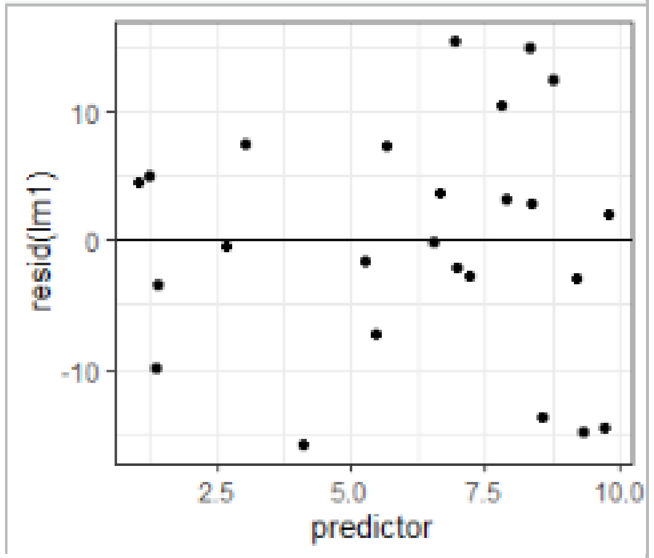


Residuals should be evenly scattered around zero when plotted against the original predictor

Assumptions of linear models

- Response variable errors (ie. residuals) normally distributed
- Linearity
- **Homoscedasticity**

Good



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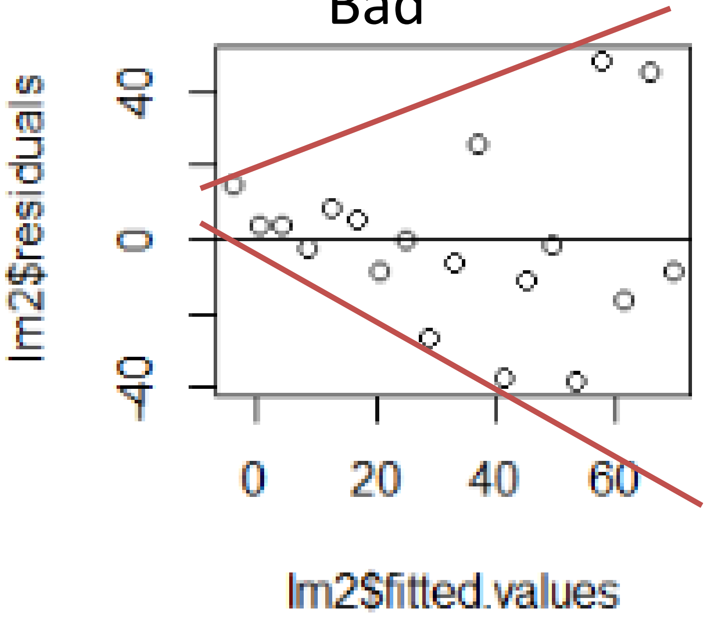
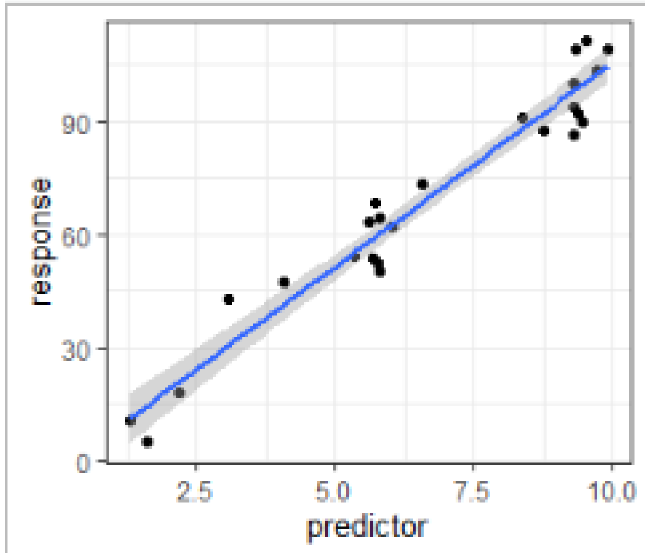


Residuals should be evenly scattered around zero when plotted against the original predictor

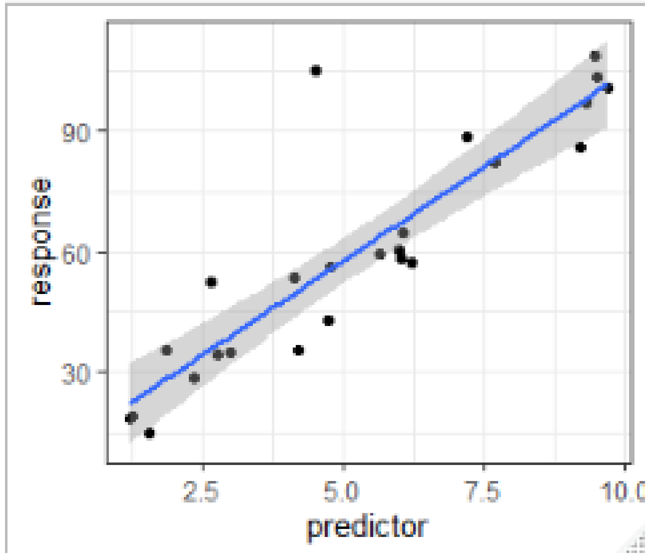
Assumptions of linear models

- Response variable errors (ie. residuals) normally distributed
- Linearity
- Homoscedasiticy
- **Outliers**

Good



Okay



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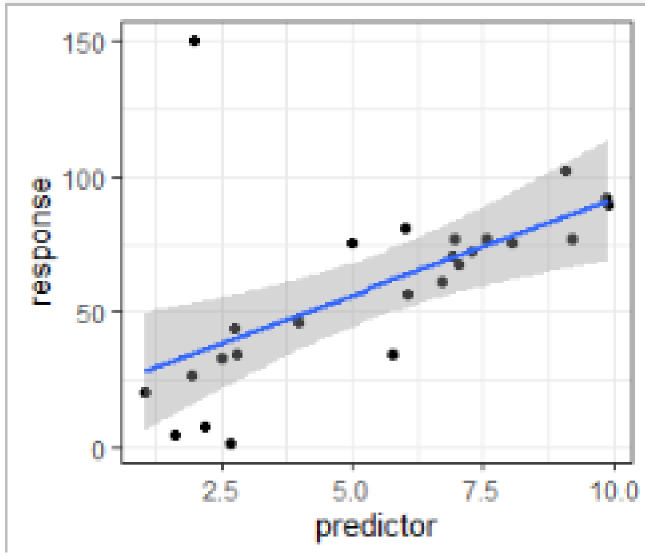
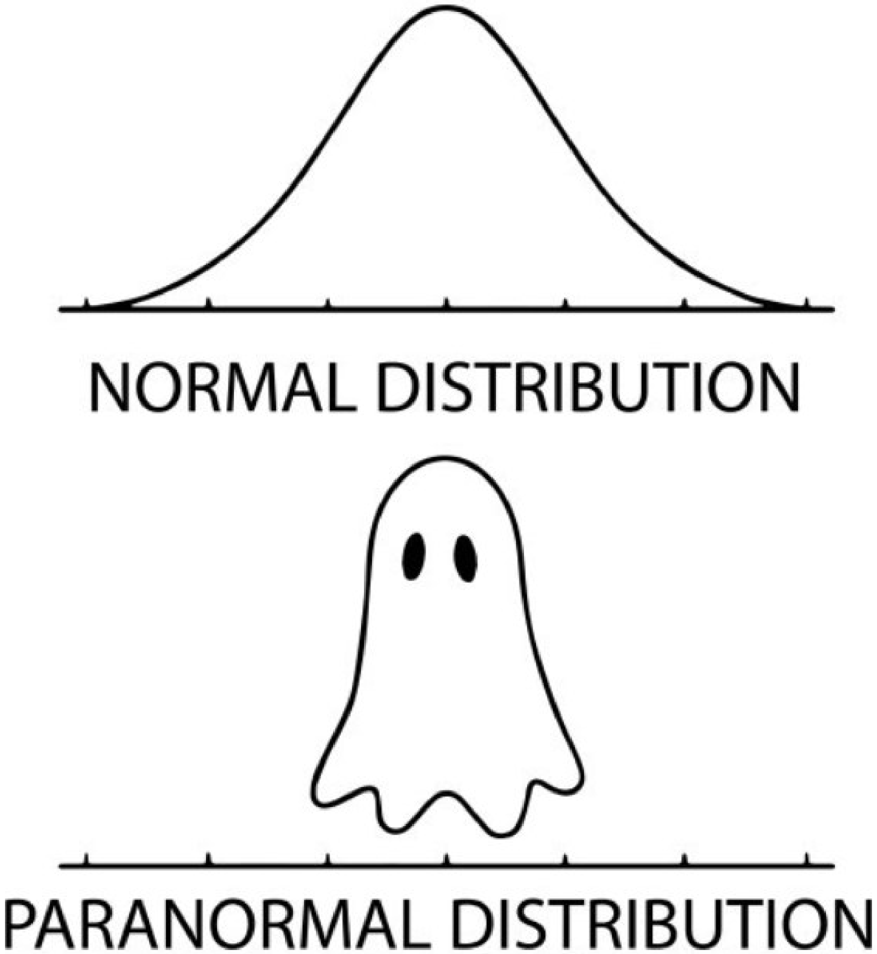


Often data (and residuals) are not normally distributed

Why might we need to use non-normal distributions?

- Linear models are *reasonably* robust to violations of the assumptions
- Violations can affect
  - P-values
  - Standard errors
  - Parameter estimates
- Generalized models provide a better fit to the data
  - P-values
  - Standard errors
  - Parameter estimates
- Reviewers and editors expect it

Questions ???



Break

Materials and code:

What are "generalized" models?

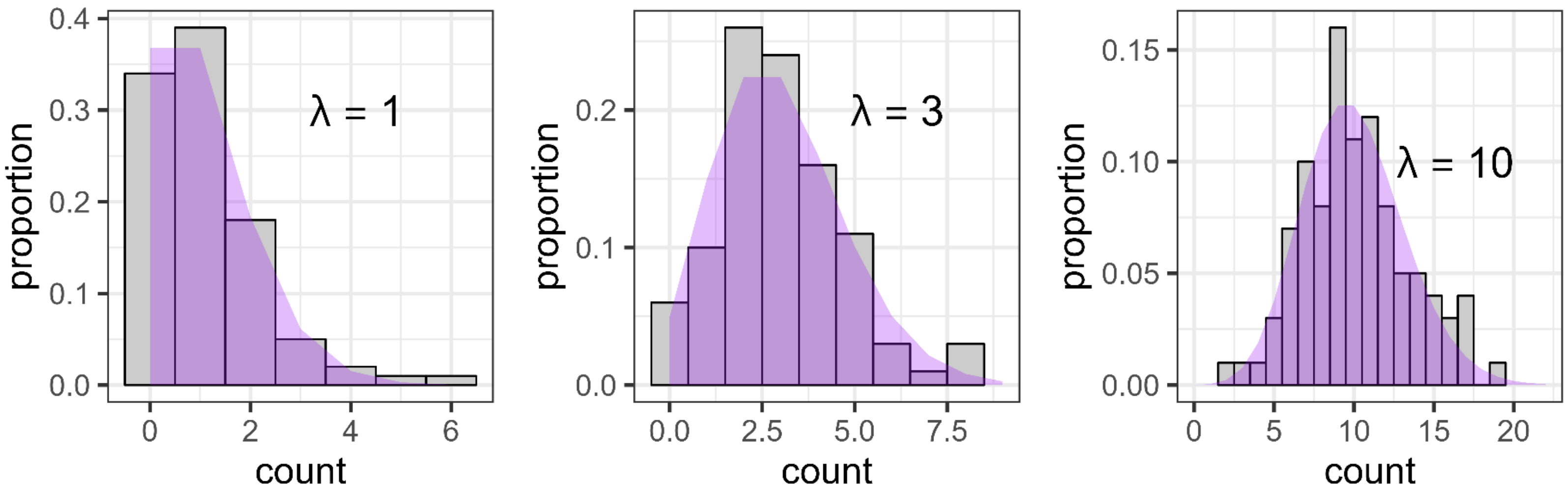
- Linear models that allow for more flexible error distributions
- Parameters estimated via maximum likelihood (not least squares)
- Model components
  - Linear predictor formula (y ~ x)
  - Error distribution (usually exponential family)
  - Link function: links distribution to linear predictor formula
    - ie. Transforms error distribution to linear
- Generalized linear **mixed** model
  - Allows fixed and random effects

Generalized linear **mixed** models

- Linear model: $y_{ij} \sim \mu + \beta_i + \varepsilon_{ij}$
- Linear mixed model: $y_{ij} \sim \mu + \alpha_i + \beta_j + \varepsilon_{ij}$
- Generalized linear mixed model: $\log(y_{ij}) \sim \mu + \alpha_i + \beta_j + \varepsilon_{ij}$

Poisson distribution - assumptions

- Discrete probability distribution
  - Integers: number of events (per unit of time or space; ie. rate)
  - Events occur randomly
  - Events are independent
- Parameters
  - λ = mean = variance
  - Link function = log



Poisson linear model

Parameters

Intercept Slope

$$\log(y_i) \sim \beta_0 + \beta_1 X_i$$

Response variable Predictor variable

Data

Poisson linear model

Parameters

Intercept Slope

Link function:

- Relates x to log(y) on linear scale
- Works on predicted values (λ_i)
- Doesn't transform raw data

$$\log(\lambda_i) \sim \beta_0 + \beta_1 X_i$$

Response variable (predicted)

Predictor variable

Data

Poisson linear model

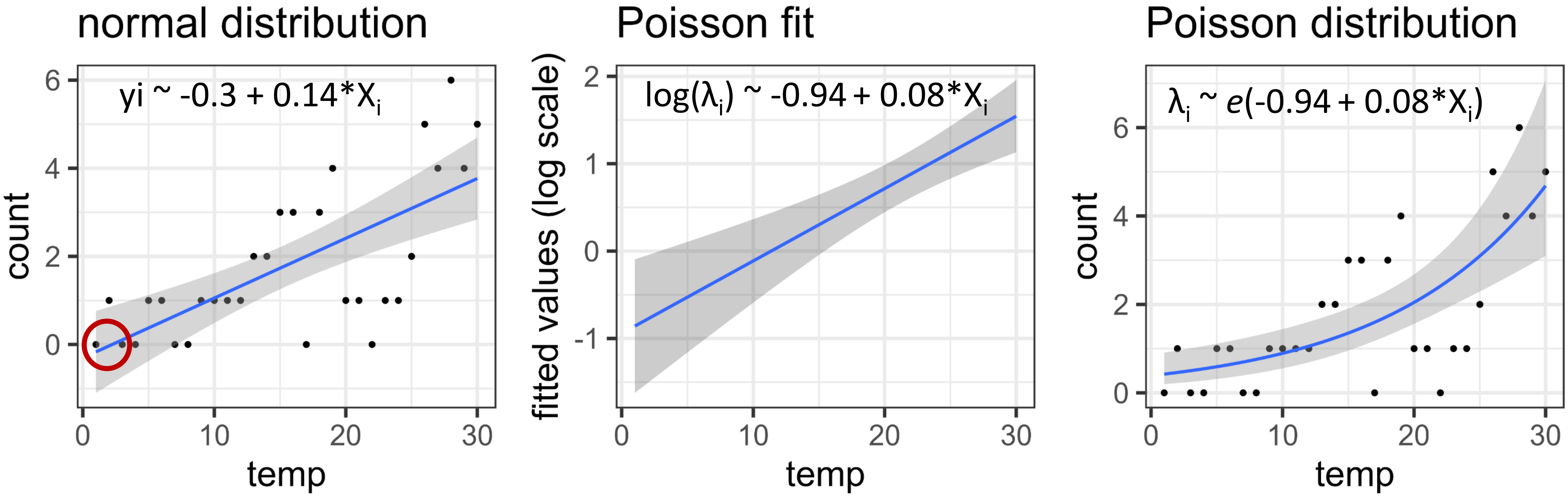
$$\log(\lambda_i) \sim \beta_0 + \beta_1 X_i$$

$$=$$

$$\lambda_i \sim e^{\beta_0 + \beta_1 X_i}$$

Poisson linear model – example 1

- Counts across temp (continuous, 1 to 30):
- $\log(\lambda_i) \sim \beta_0 + \beta_1 X_i$



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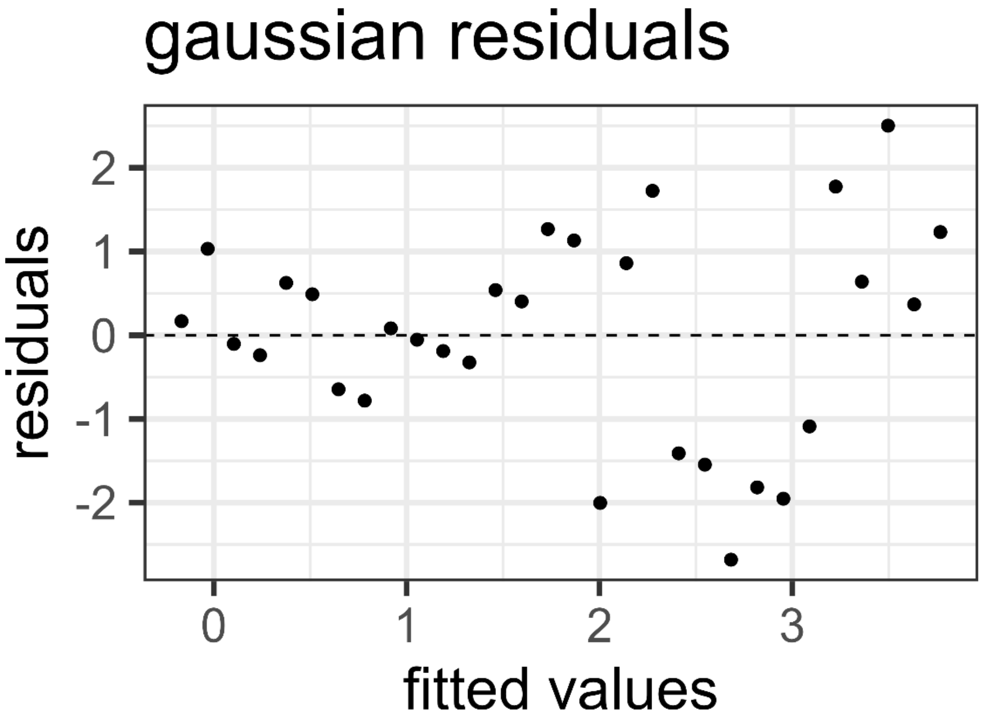
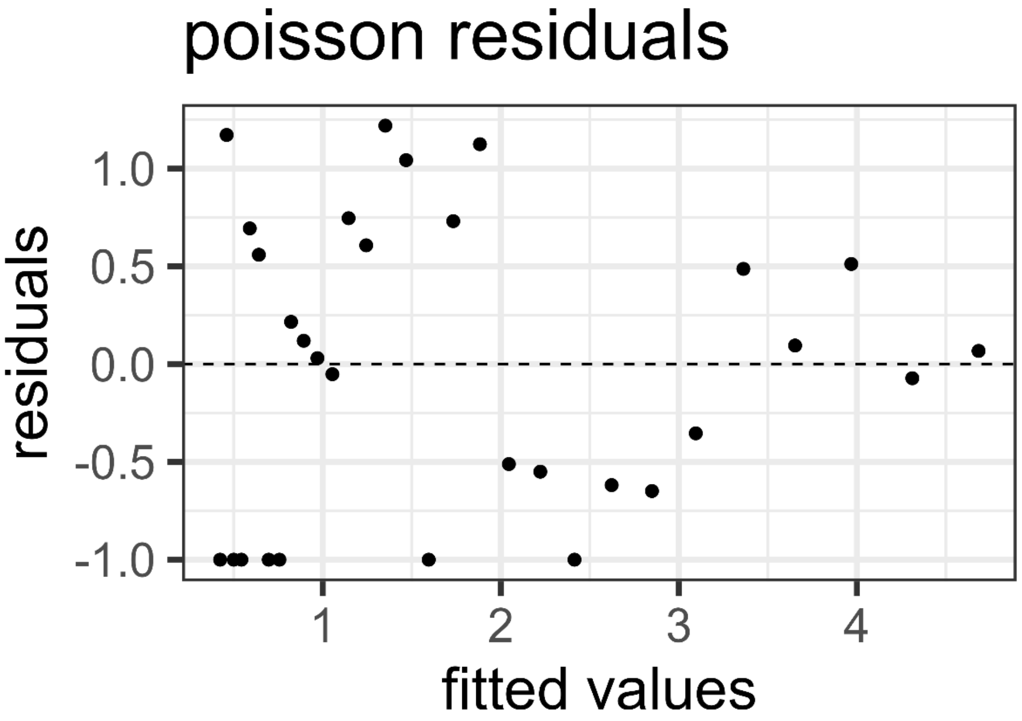
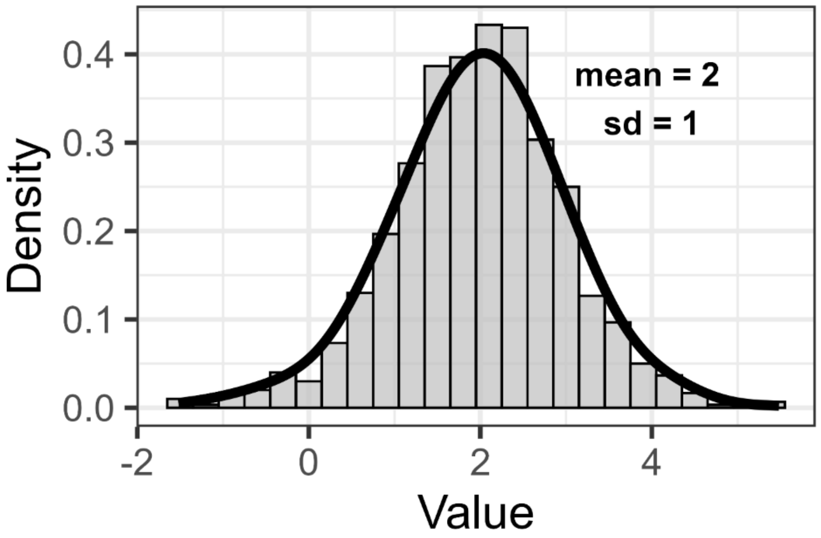




Common distributions

Normal (gaussian)

- Continuous numeric
- Parameters
  - Mean
  - Variance (SD)
- Link = identity (none)



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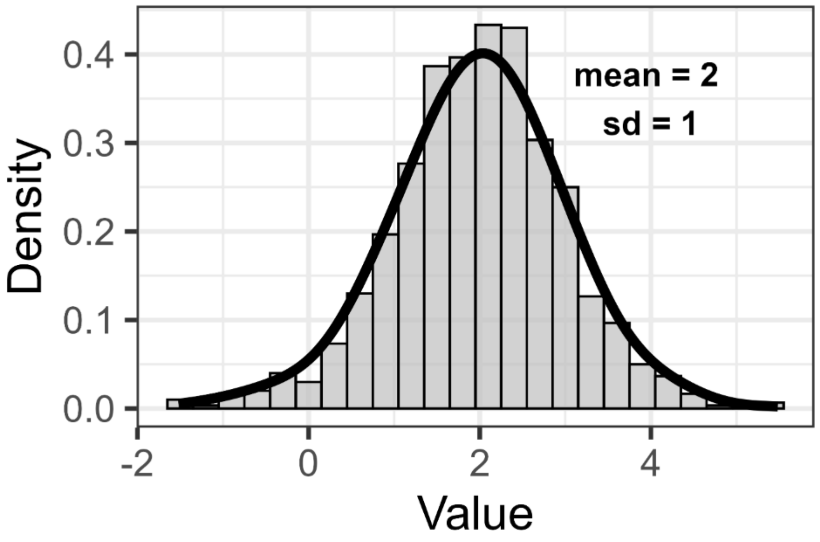
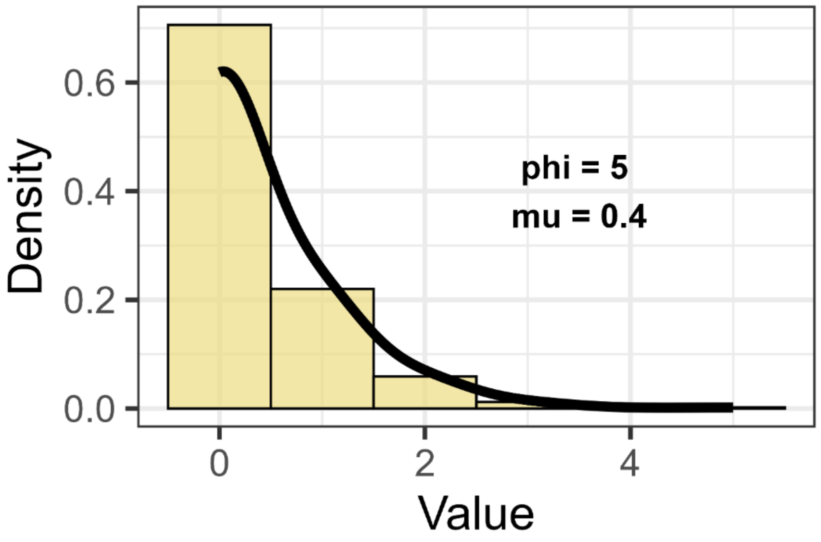


Negative binomial

- Counts
- Parameters
  - mu (mean)
  - phi (precision)
    - Inverse of dispersion
- Link = log



Common distributions

Normal (gaussian)
- Continuous numeric
- Parameters
  - – Mean
  - – Variance (SD)
- Link = identity (none)

Negative binomial
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  - phi (precision)
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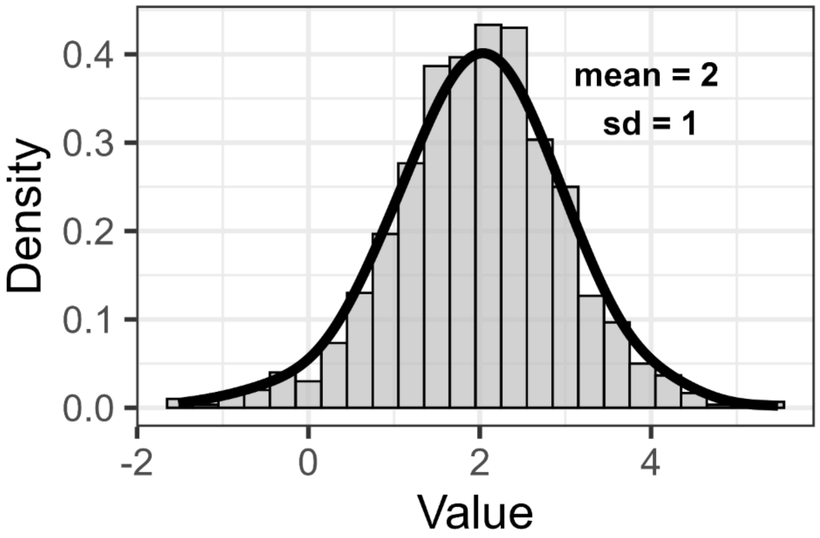
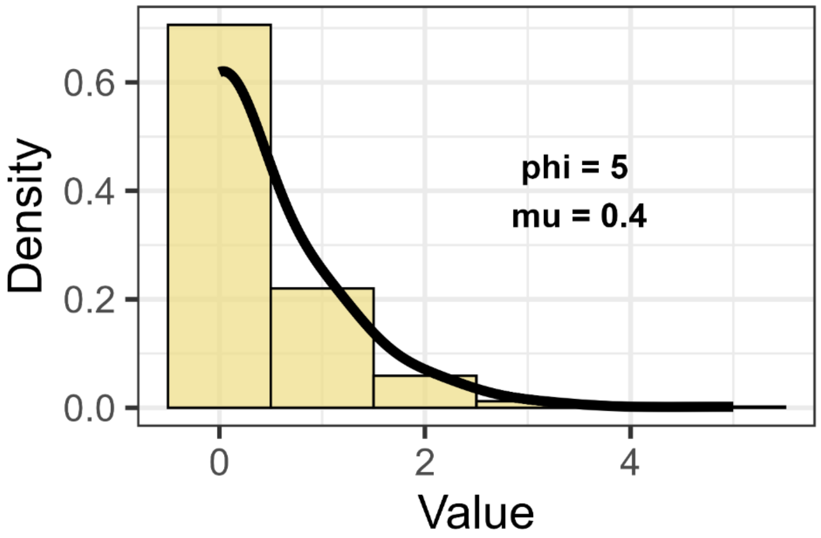
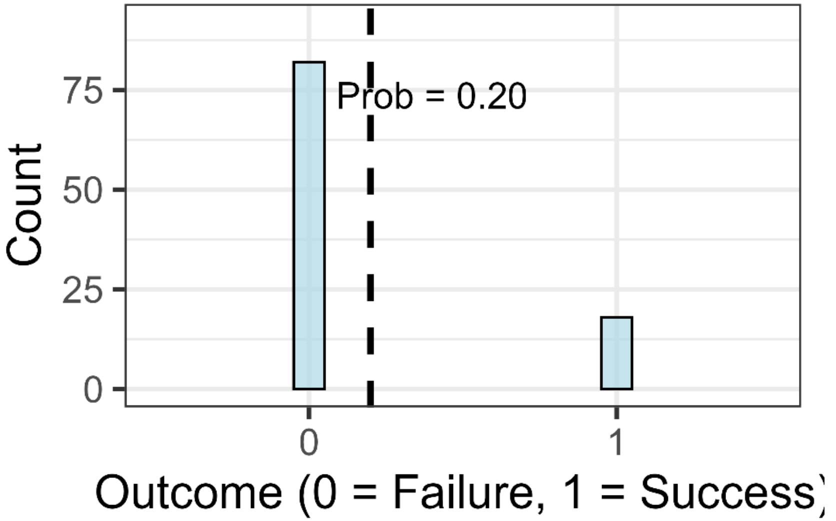
- Link = log

Binomial
- Proportions
  - – **<u>Binary</u>** or ratio
- Parameters
  - – Probability
  - – # trials
- Link = logit







Common distributions

Normal (gaussian)

- Continuous numeric
- Parameters
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Negative binomial

- Counts
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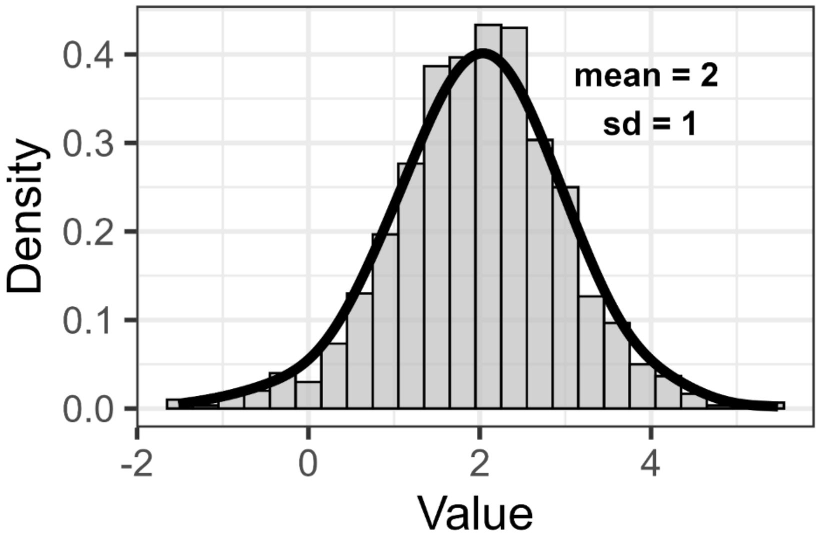
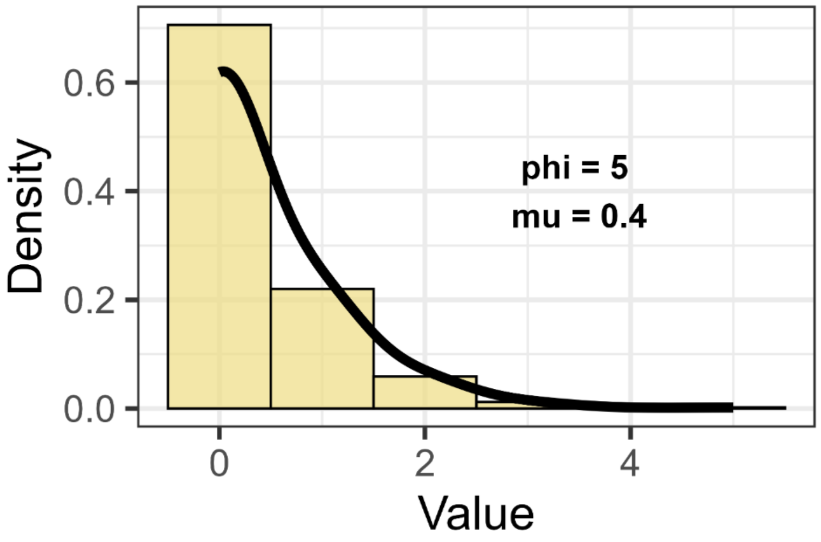
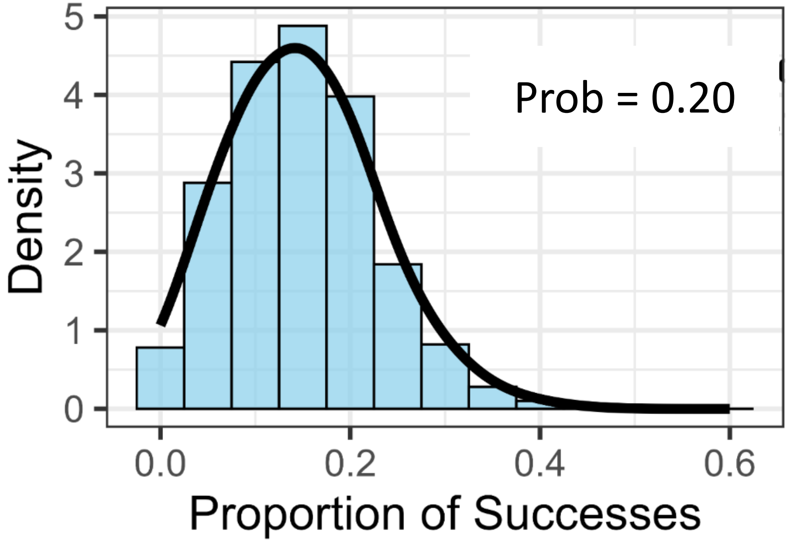
- Link = log

Binomial

- Proportions
  - Binary or **ratio**
- Parameters
  - Probability
  - # trials
- Link = logit

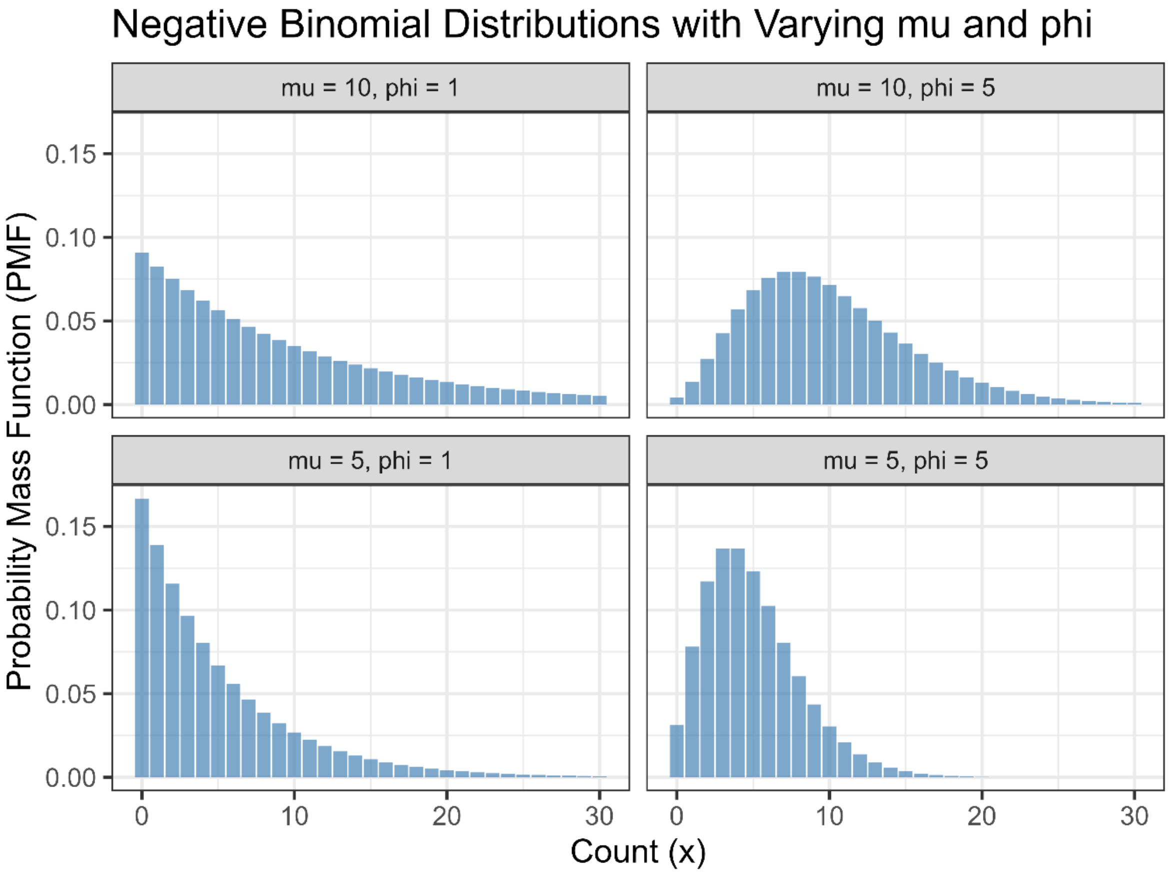






Negative binomial distribution - assumptions

- Discrete probability distribution
  - Integers: number of events
  - Events occur randomly (follow Poisson distribution)
  - Account for "extra" variation (beyond poisson)
- Parameters
  - μ = mean
  - ϕ = overdispersion parameter
    - lower values are more overdispersion
    - higher values converge to Poisson
  - $Var = \mu + \mu^2 / \phi$
  - Link function = log



Common types of distributions

Distribution	Type of data	Overdisperion parameter?	Link
Gaussian	Continuous numeric variable	Yes	Identity
Log-normal	Continuous numeric variable	Yes	Identity
Poisson	Integers, counts	No	Natural log
Negative binomial	Integers, counts	Yes	Natural log
Binomial	Success/failure or presence/absence (binary), ratio	No	Logit, cloglog
Beta-binomial	Success/failure, presence/absence (binary), ratio	Yes	Logit, cloglog
Beta	Proportion (when the denominator is not known)	Yes	Logit
ordbeta	Proportion (when the denominator is not known), can have zero’s	Yes	Logit
Gamma	Positive continuous variable	Yes	Inverse, log
ziGamma	Positive continuous variable, can haze zero’s	Yes	log

What distribution best matches the data?

Description of data

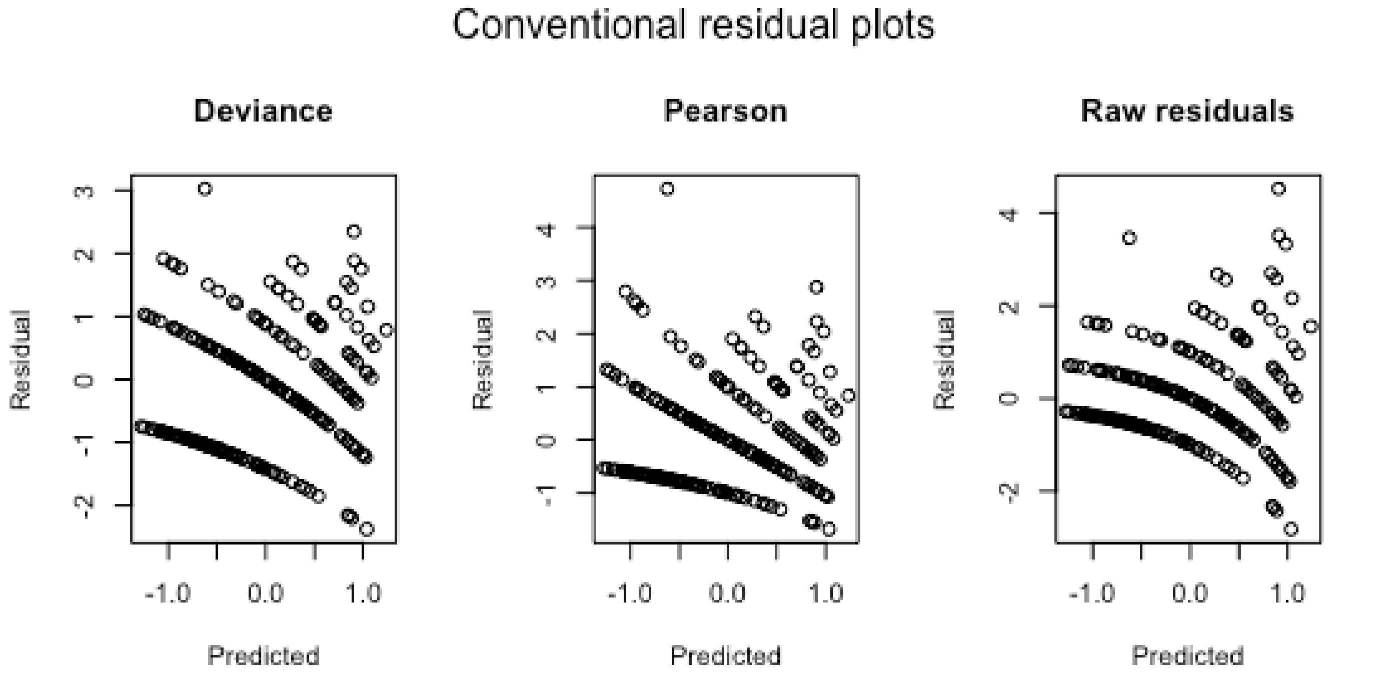
1. Concentrations of chemicals in the soil, most values are low (~0.1 ug/L) but some are high (5-10 ug/L)
2. Number of infected and uninfected aphids on plants
3. Percent damage on leaves caused by herbivore or pathogens
4. Number of thrips on a plants

Distribution

A. Normal
B. Log-normal
C. Poisson
D. Negative binomial
E. Binomial
F. Beta
G. Gamma

Step 2. Check GLM assumptions

- "Conventional" residuals don't work well for non-normal data



This figure shows residuals fit for count data assuming a Poisson distribution.
The model (not shown) fits the data well, although the residuals look wonky.

Step 2. Check GLM assumptions

- “Regular” residuals often don’t work well for non-normal data
- Simulated residuals offer a great diagnostic tool

DHARMa: residual diagnostics for hierarchical (multi-level/mixed) regression models

Florian Hartig, Theoretical Ecology, University of Regensburg website

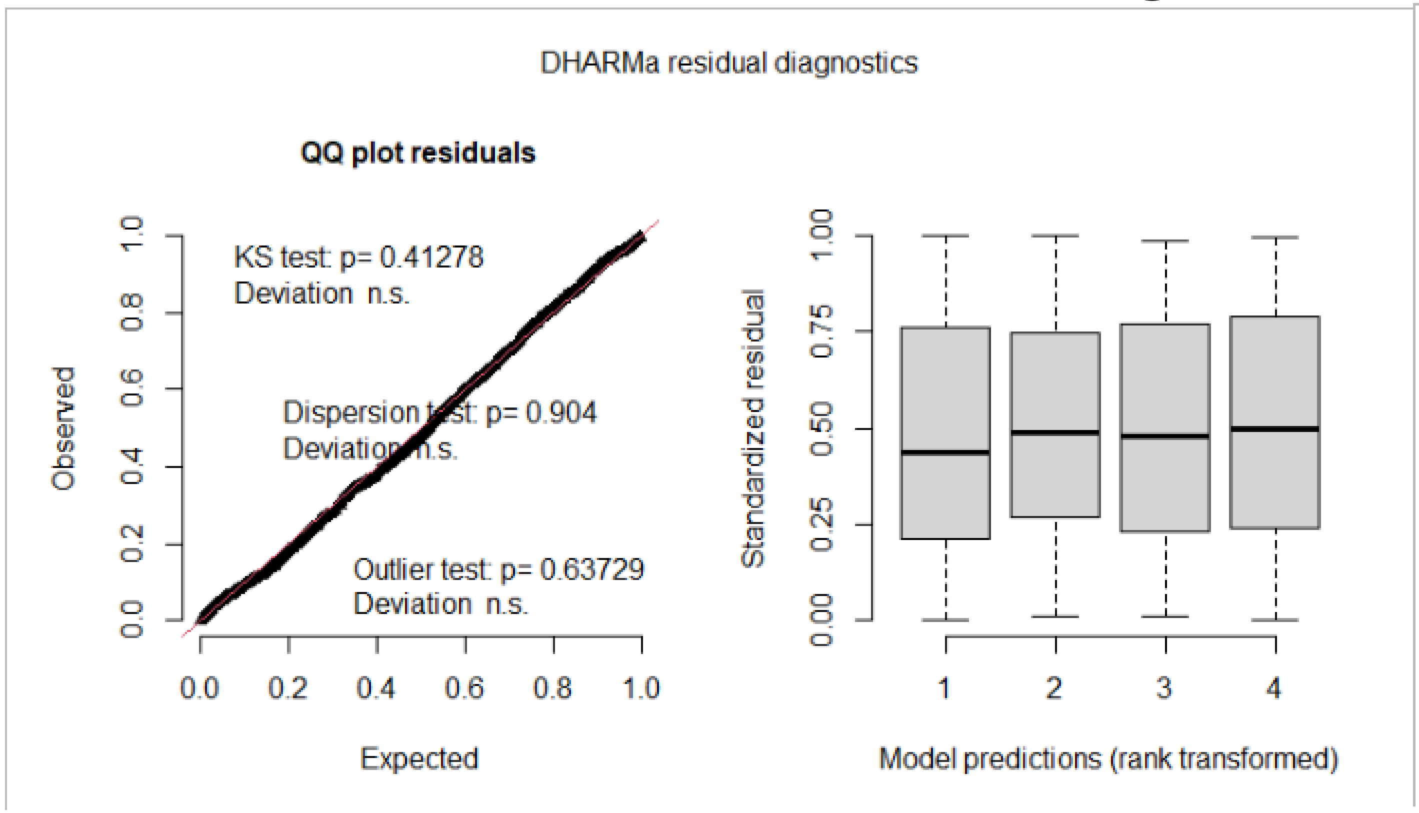
2021-03-27

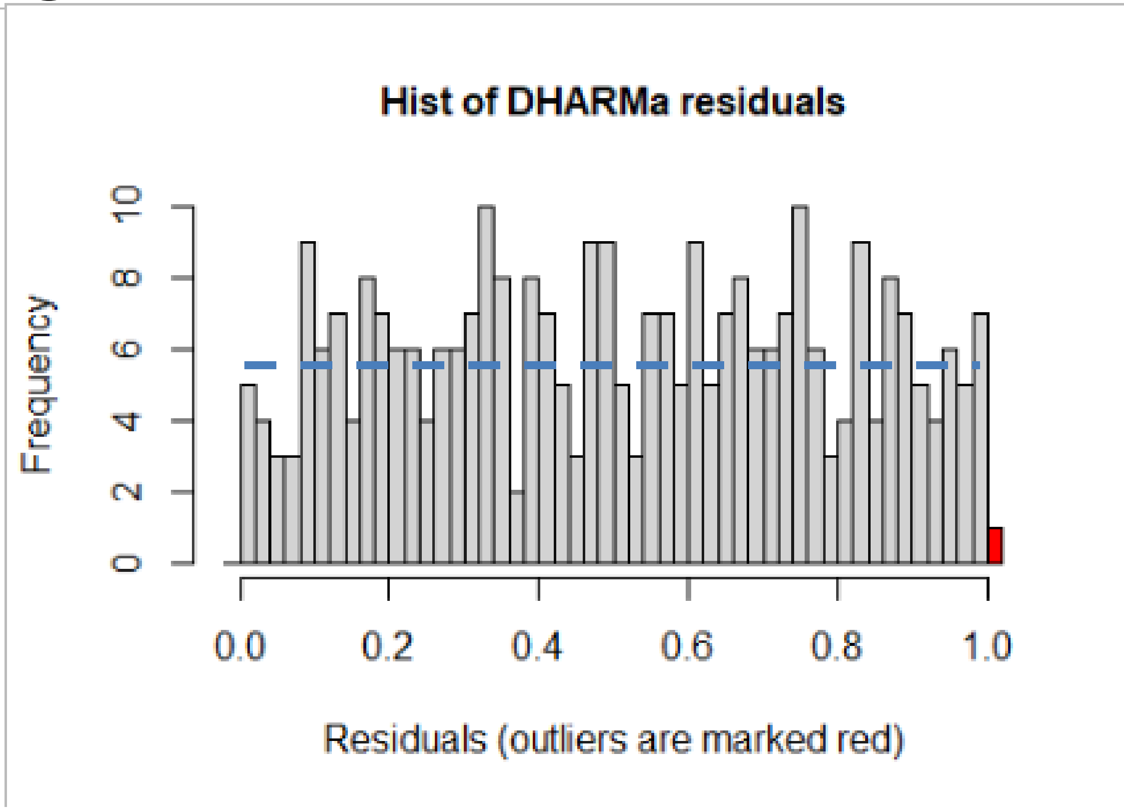
DHARMa stands for “Diagnostics for HierArchical Regression Models”

https://cran.r-project.org/web/packages/DHARMa/vignettes/DHARMa.html

Step 2. Check GLM assumptions

- “Regular” residuals often don’t work well for non-normal data
- Simulated residuals offer a great diagnostic tool





https://cran.r-project.org/web/packages/DHARMa/vignettes/DHARMa.html

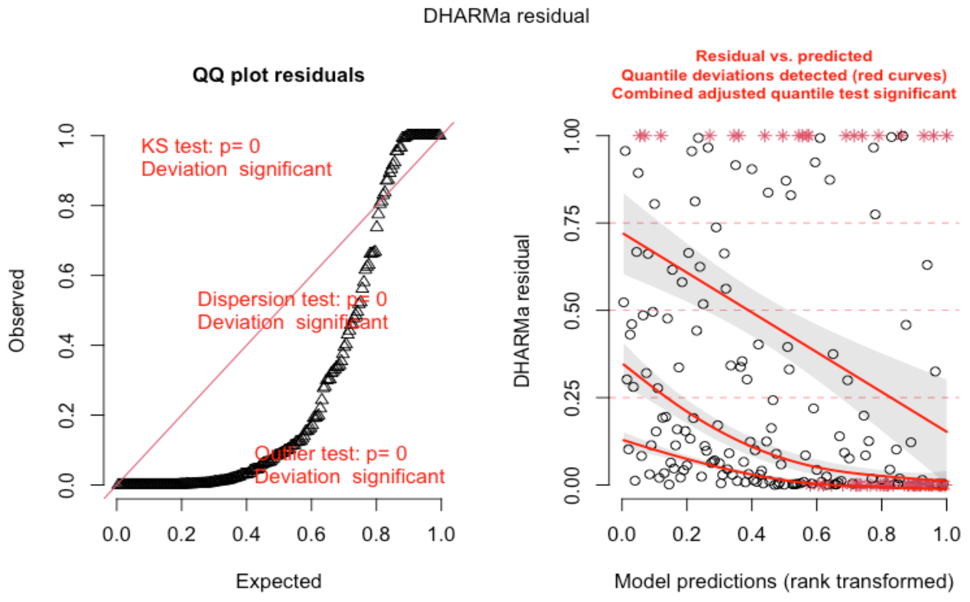
Reasons for deviations of residuals

- Mis-specified model
  - Incorrect distribution (or potentially link function)
  - Missing fixed effect
    - Include additional covariates, if available
    - Check outliners in covariates or influential data points
    - Include quadratic term
  - Incorrect random-effect structure
    - Missing block or nestedness?
    - Incorrect order of nesting?
- **Overdispersion** (return to this later)
- Zero-inflation

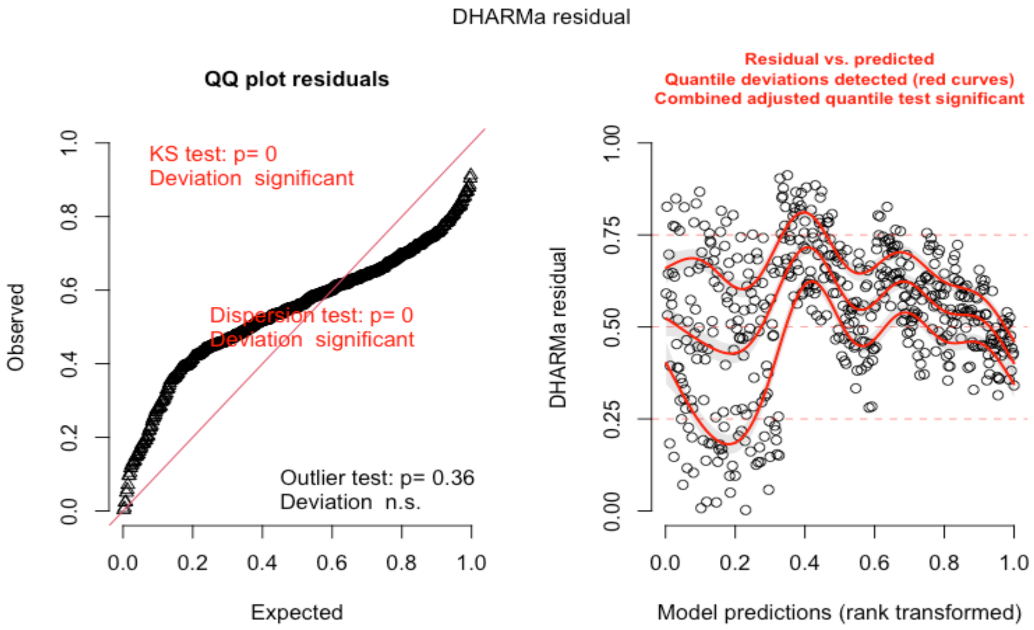
Step-by-step guide to analysis (post-data collection)

2. Check model assumptions

Bad residuals (overdispersion)



Bad residuals (wrong dist., overfitting, etc.)



Step 3. Check random effects and model components

Investigate model with summary()

- Random effects
  - Examine variance components
- Fixed effects
  - Check fixed-effect signs, magnitudes, and uncertainty
  - Consider whether all predictors and interactions remain justified
- Fit metrics (e.g., AIC)
  - Compare a set of plausible models

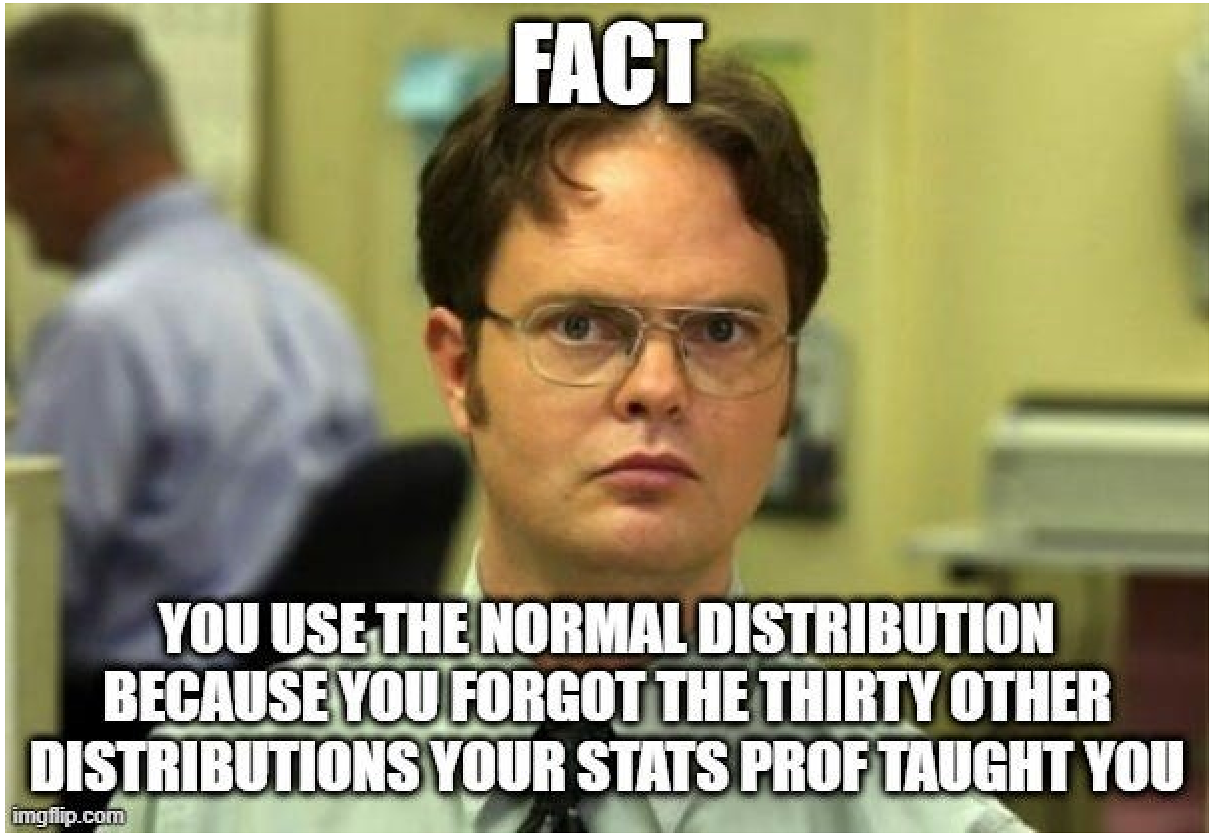
Check random effects

- Not always necessary
  - Not all models have/need random effects and non-normal distributions
- Very small/zero variance components suggest simplification
  - Singularity can be worth noting (very small VC is *almost* singular)

AIC for model comparison

- smaller AIC = better fit–complexity trade-off

Questions ???



Materials and code:

Step 4. Check significance of terms in model

Where do p-values come from?

- Estimate parameter values (i.e., slopes or intercepts)
- Calculate a test-statistic
- Use a distribution to find your p-value

Alternative method is AIC model selection

Methods to estimate parameters

Least squares

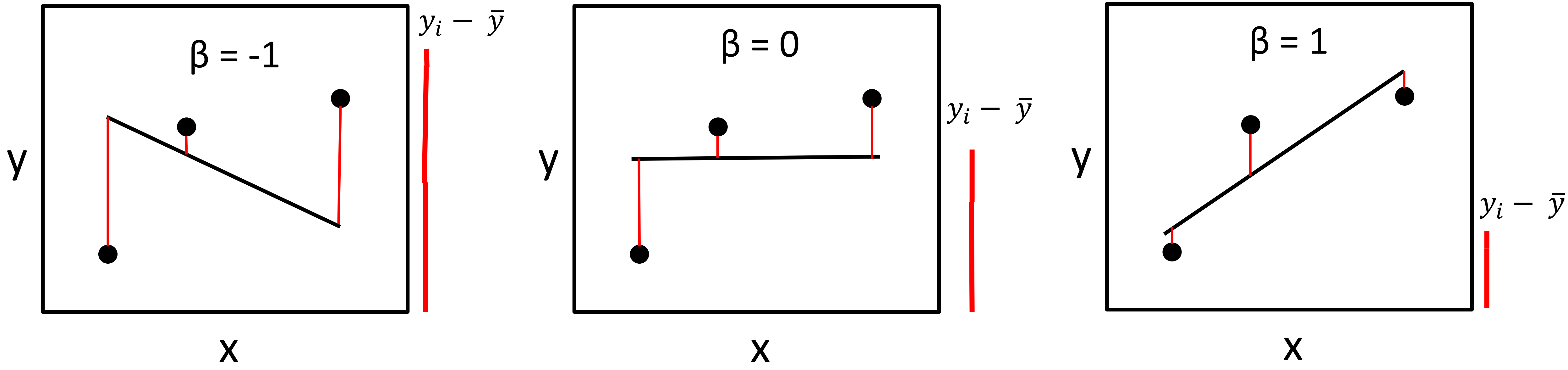
- Minimizes the sums of squares
- $\sum(y_i - \bar{y})^2$

Maximum likelihood

- Maximizes the likelihood function
- $\mathcal{L}(\beta|y)$

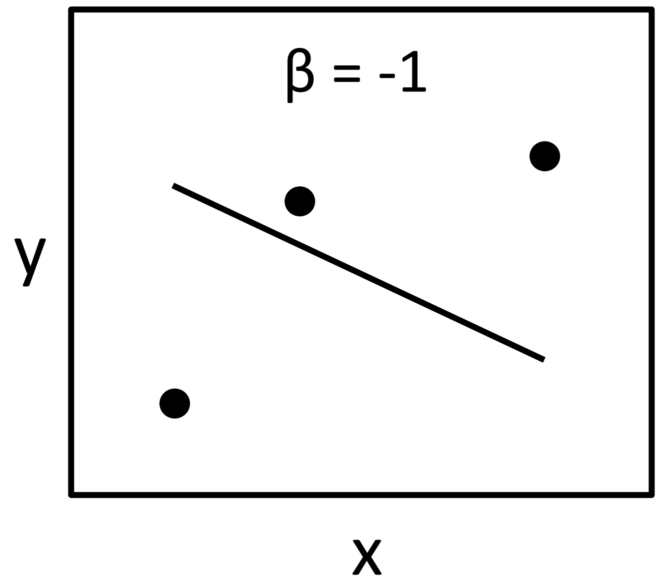
Least squares estimation

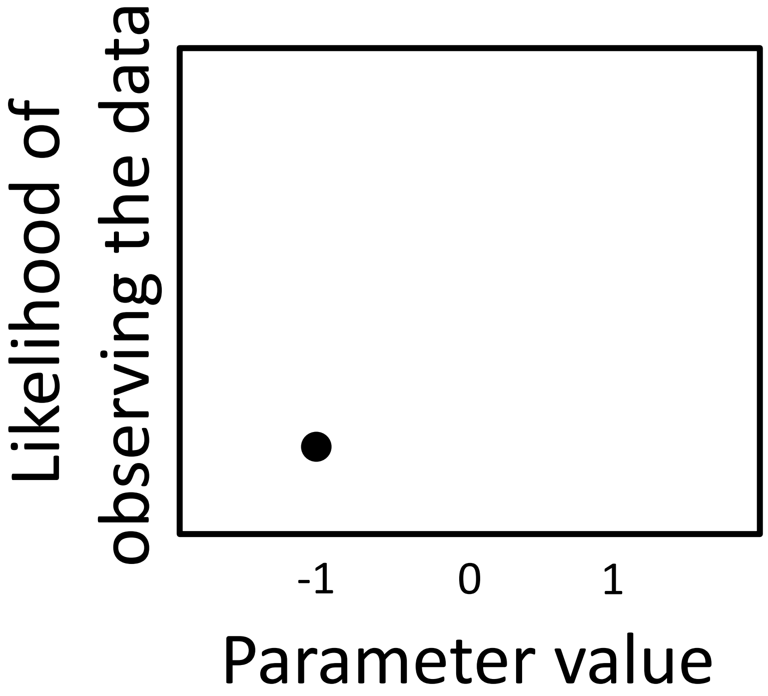
- Minimizes the sums of squares
- $\sum(y_i - \bar{y})^2$
- β = slope; Fit represents $\bar{y}$



Maximum likelihood estimation

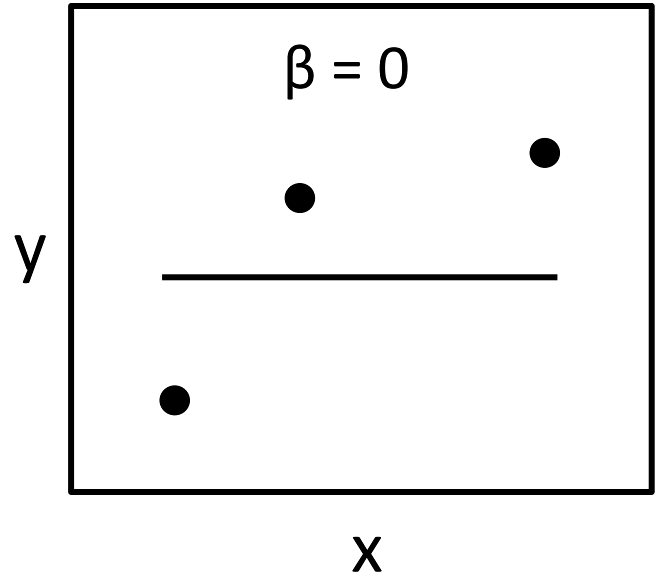
- Maximizes the likelihood function
- $\mathcal{L}(\beta|y)$: How likely is β given the observed data (y)
- β = slope

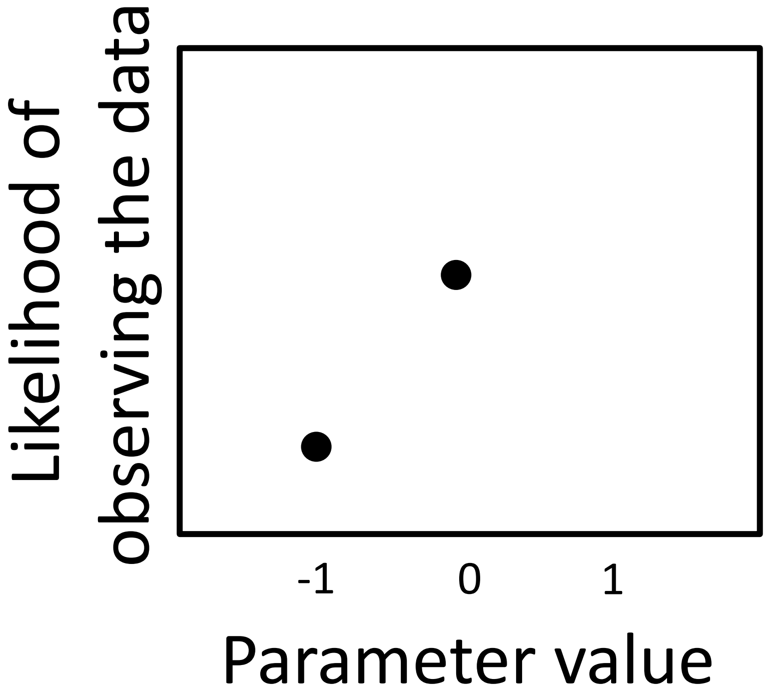




Maximum likelihood estimation

- Maximizes the likelihood function
- $\mathcal{L}(\beta|y)$: How likely (probable) is β given the observed data (y)
- β = slope





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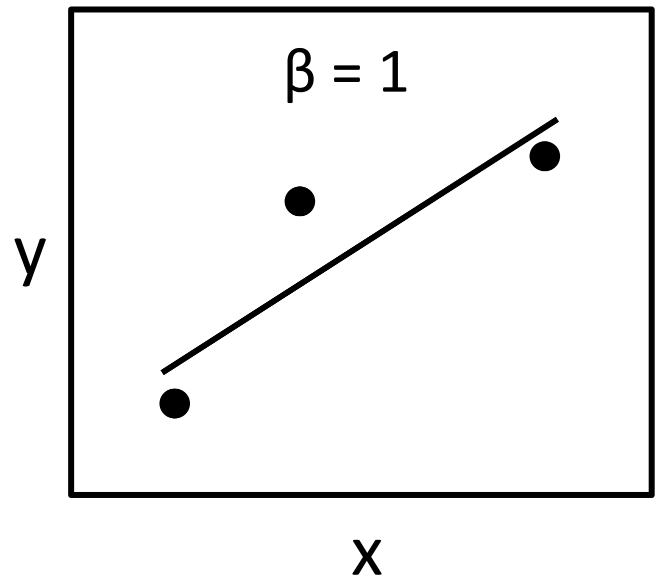

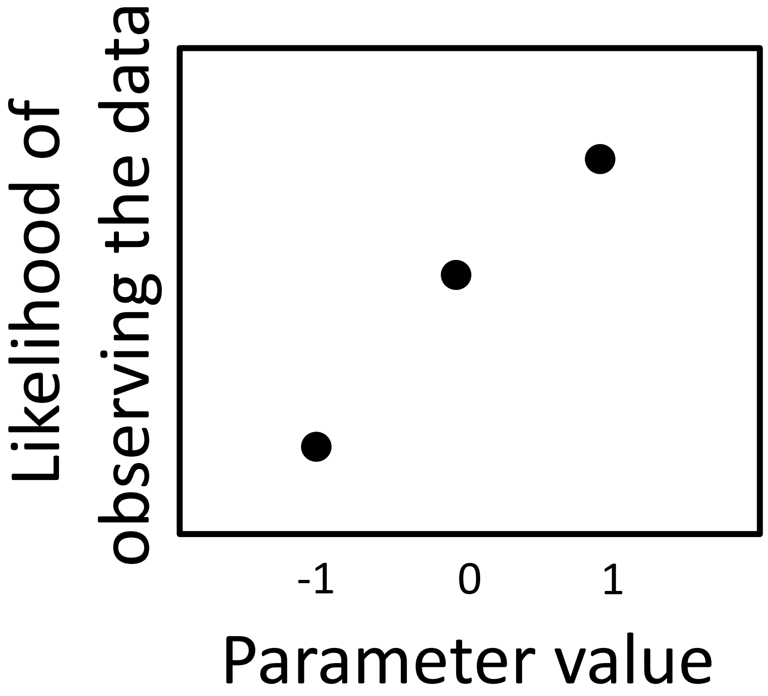

Maximum likelihood estimation

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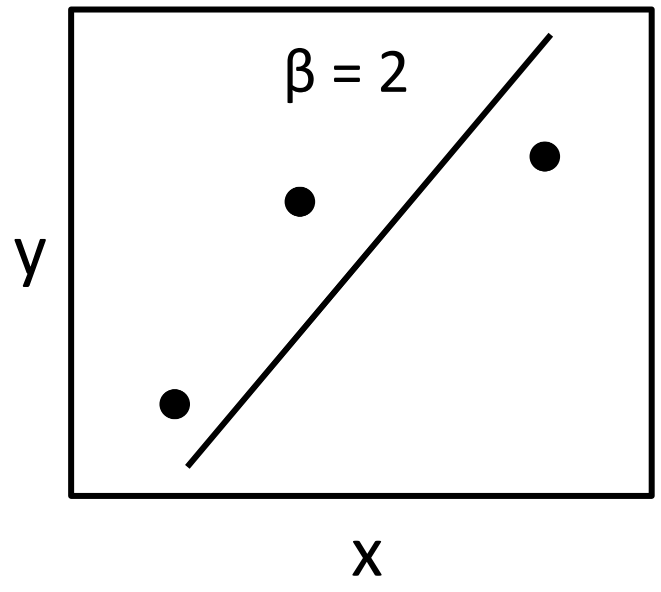
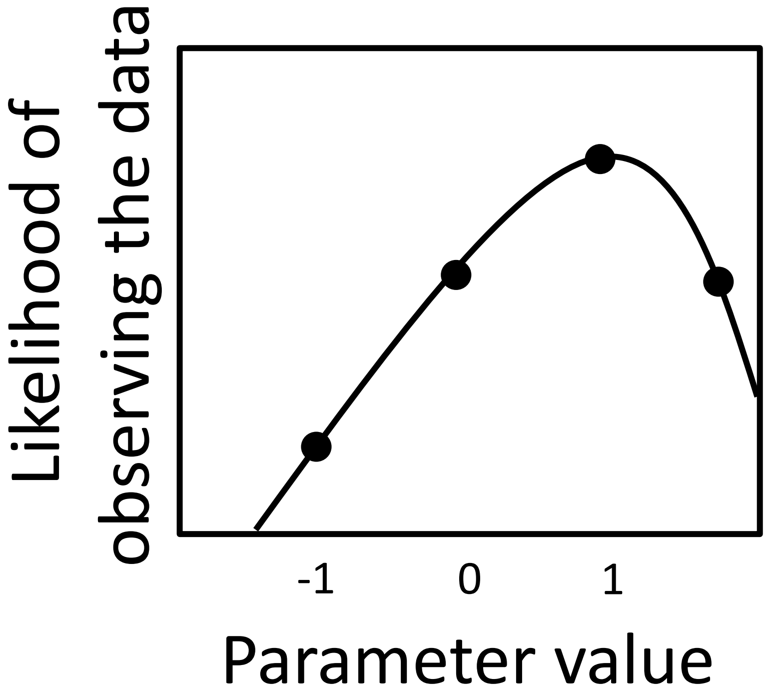
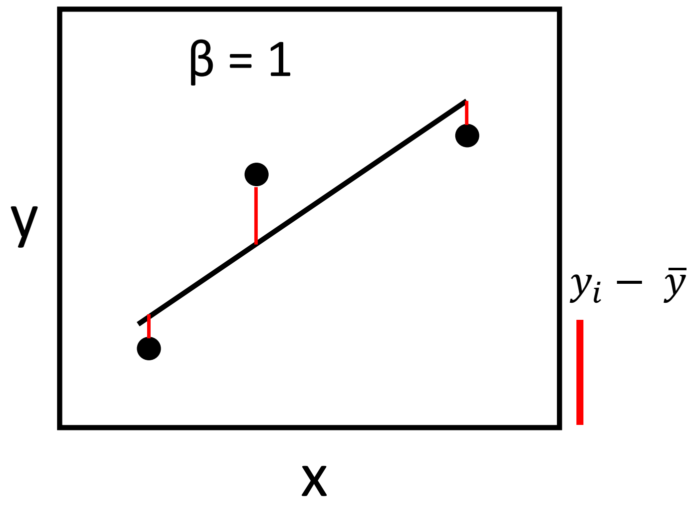




Methods to estimate parameters

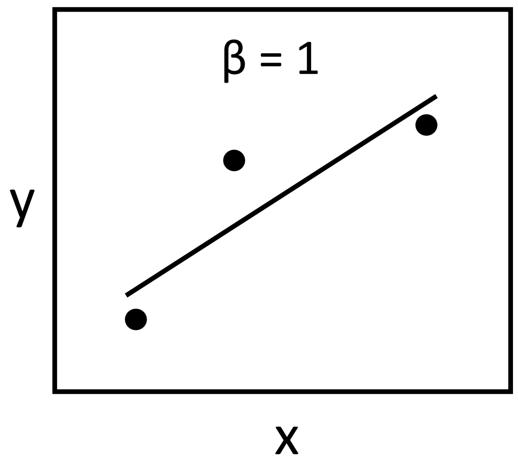
Least squares

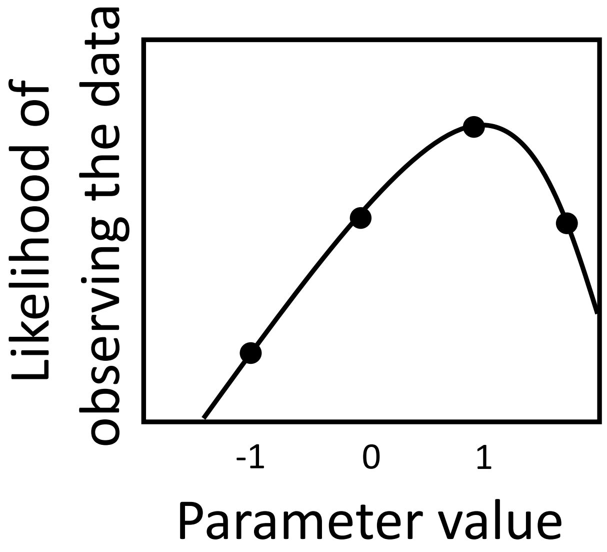
- Minimizes the sums of squares
- $\sum(y_i - \bar{y})^2$
- β = slope or intercept adjust
- Only works for normal data



Maximum likelihood

- Maximizes the likelihood function
- $\mathcal{L}(\beta|y)$
- β = slope or intercept adjust
- Can be generalized to any distribution

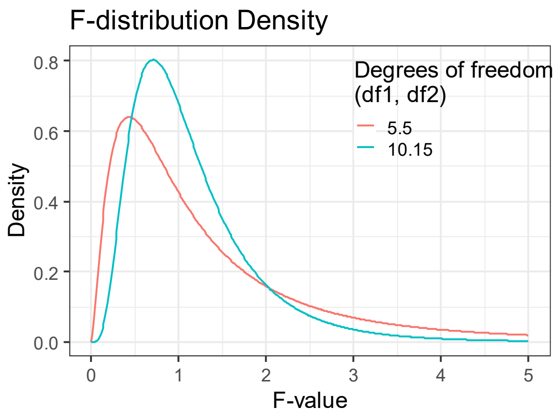




Methods to estimate parameters

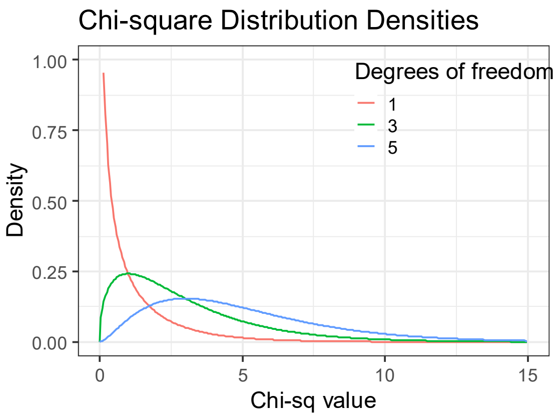
Least squares

- Minimizes the sums of squares
- $\sum(y_i - \bar{y})^2$
- β = slope or intercept adjust
- Only works for normal data
- F-value for significance test
  - F = MS_B / MS_W



Maximum likelihood

- Maximizes the likelihood function
- $\mathcal{L}(\beta|y)$
- β = slope or intercept adjust
- **Can be generalized to any distribution**
- χ2 for significance test
  - χ2 = (effect size / SE)^2



After break

- Worked example
- Overview Steps 1-4
- **Step 5: Calculate group means**
- **Step 6: Conduct contrasts**
- **Step 7: Calculate R2**
- Step 8: Visualize data (if time permits)

Materials and code:

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